

Who Holds U.S. Bonds and How It Shapes the Yield Effects of QE and QT

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Abstract

How much do the bond-supply shifts of quantitative easing (QE) and tightening (QT) move yields? The answer turns on the slope of aggregate bond demand: who holds bonds, how sensitive their demand is to yields, and how fast they re-balance. I estimate heterogeneous investor demand in the two U.S. bond markets the Federal Reserve trades, Treasuries and agency MBS, from sector-level portfolio holdings and transactions spanning 1985 to 2025. I measure each sector's demand against its own non-bond alternative and let its yield sensitivity and rebalancing speed vary. Most investors respond to yields weakly and slowly, so aggregate demand is inelastic. Feeding the demand system through market clearing, I find that historical QE compressed 10-year Treasury yields by 8 to 12 basis points per percent of the float removed. Because agency MBS demand prices the spread over Treasuries, the composition of a program, not its size, determines what it does to mortgage rates. A further runoff would move yields less per unit than the QE programs did, because the sectors that absorb Treasuries hold a larger share of the float than they did during QE: a further \$1 trillion of runoff raises 10-year Treasury yields by about 22 basis points and agency MBS yields by about 27, of which 5 is the spread.

Keywords: Agency MBS, Demand System, Institutional Investors, Market Macrostructure, Quantitative Easing, Quantitative Tightening, U.S. Treasuries

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1 Introduction

For over a decade, the Federal Reserve has purchased more than \$6 trillion of U.S. Treasuries and agency MBS to lower yields, a policy known as quantitative easing (QE). It has since reversed course, shrinking its balance sheet through quantitative tightening (QT), yet the consequences for bond yields and the channel through which they operate remain unclear.¹ The key is that QE and QT shift the supply of bonds available to private investors, so their effect on yields depends on the slope of investor demand curves: flat demand implies small yield effects, steep demand large ones (e.g., [Kojien and Yogo, 2019](#); [Vayanos and Vila, 2021](#)). Because investors differ in their yield sensitivity, in the assets they hold, and in how quickly they rebalance, the aggregate demand curve, and hence the size and timing of the yield effect of QT, depends on *which* investors absorb the released supply, and how fast.

This paper estimates bond demand using sector-level portfolio holdings and transactions covering the two bond markets the Federal Reserve trades, Treasury securities and agency MBS, across 26 sectors grouped into eight broad investor types: banks, insurers, pension funds, mutual funds, arbitrageurs, foreign investors, households, and other investors. Four choices define the system. I measure each sector’s demand from its transactions rather than from the market value of its holdings. I measure it against the non-bond asset each sector actually trades off against bonds, its own outside asset. I let each sector’s yield sensitivity differ by asset class, so the system can say who absorbs Treasuries and who absorbs agency MBS instead of imposing one slope per investor across markets. And I let each sector rebalance at its own speed, estimating how far and how fast each adjusts. The estimates support counterfactual analyses of the QE and QT episodes and of the runoff to come.

I find three main results. First, most investors are inelastic, and the few that absorb supply do so slowly. In the Treasury market only two sectors buy when the yield rises: the household residual, which after 2010 contains the hedge funds running the cash–futures basis

¹In June 2026, Federal Reserve Chairman Kevin Warsh made balance-sheet policy one of five priorities for the conduct of monetary policy, charging a dedicated task force to weigh “the benefits and risks of the current ample reserves regime, and the composition of the Fed’s balance sheet” ([Warsh, 2026](#)).

trade, and mutual funds. Banks, insurers, pension funds, and foreign investors, who hold most of the float, do not move. In the agency MBS market, by contrast, demand responds to the option-adjusted spread rather than to the yield level, and every sector type absorbs. The data reject the restriction, standard in bond demand systems, that an investor’s slope is the same across the bonds it holds. Most absorbers complete only an eighth to a quarter of their one-year response in the quarter of the shock. Second, because the Treasury market’s absorbers are few, Treasury yields are highly sensitive to supply: historical QE compressed 10-year Treasury yields by 8 to 12 basis points for each percent of the float the Fed removed, from 54 basis points under QE1 to 186 under QE4, in line with the demand-system literature (Eren et al., 2026; Chaudhary et al., 2026). Because the Treasury market sets the level of mortgage yields and the MBS market only the spread on top, Treasury purchases lower mortgage yields too: QE2, a Treasury program, lowered them by 90 basis points, and QE1, the Fed’s only mortgage-heavy program, compressed the spread by a further 54. Third, a resumed QT raises mortgage costs mostly through the Treasury market. A further \$1 trillion of runoff from the 2025Q1 balance sheet, about a year and a half at the Fed’s 2024 pace, widens the mortgage spread by only 5 basis points but raises 10-year Treasury yields by 22, so mortgage yields rise by 27. The Treasury number is uncertain: it ranges from 6 basis points in a recession to 45 at the lower edge of the household residual’s confidence set, because the two sector slopes it depends on, the household residual’s and the foreign sector’s, are not pinned down: the first is imprecise, and the second depends on which instrument identifies it.

I construct quarterly sector-level portfolio shares and transactions for the two bond classes from the Federal Reserve’s Financial Accounts (Z.1), splitting the rest of the world into official and private holders with the Federal Reserve’s CSLT cross-border positions data over the full sample. The Financial Accounts report every holding together with the net purchases that changed it, decomposing each level change into transactions, revaluations, and other volume changes. Like Eren et al. (2026), I separate trading from the revaluation

of existing positions; where they approximate the revaluation with a duration-based price change, I use the revaluation each sector records in the Financial Accounts. The two coincide for holders that mark to market, and the recorded flows capture the different accounting of banks, which carry held-to-maturity securities at amortized cost, and insurers, which use book value. The same flows keep the issuance characteristic free of revaluation. I merge the transaction panel with asset characteristics and macroeconomic controls from FRED. The yield is the 10-year Treasury yield for Treasuries and, for agency MBS, the 10-year yield plus the index option-adjusted spread (OAS), whose spread component is the price MBS demand responds to; maturity, duration, and coupon come from the Bloomberg indices, and issuance from the Financial Accounts. The sample covers 26 sectors and two asset classes over the 161 quarters from 1985Q1 to 2025Q1.

Given these data, I estimate a characteristics-based demand system following [Kojien and Yogo \(2019\)](#), in which each sector’s portfolio shares take a logit form, and extend it in four ways. First, the yield coefficient differs by asset, the alternative-specific-coefficient logit of discrete choice; the share structure and its closed-form elasticities are unchanged, and a minimum-distance test rejects the equal-slope restriction ([Appendix D.8](#)). Second, I measure demand from transactions rather than from market values, which keeps the revaluation of positions never traded out of the slopes. Third, because sectors rebalance at different speeds, I estimate the demand equation in first differences of transaction growth and trace the response horizon by horizon with local projections ([Jordà, 2005](#)), recovering each sector’s impact slope, long-run slope, and speed of adjustment. Fourth, the outside asset is sector-specific, loans for banks, corporate equities for long-term investors, and repo for money funds and dealers, so the slope captures substitution between bonds and each holder’s realistic alternative. The slope of each market’s aggregate demand curve, the float-weighted sum of the holders’ own-asset slopes, aggregates across sectors whatever outside asset each uses, and it is the object the counterfactual requires.

Yields are endogenous, and I identify each asset’s slope with instruments specific to

that asset. For both markets the Fed trades I use the revisions to its *announced* purchase path rather than its realized purchases, since the realized flow follows a schedule published in advance and carries no surprise. The instrument moves the mortgage spread while leaving the Treasury yield untouched, the test an asset-specific supply shifter has to pass. For Treasury yields I add a heteroskedasticity-based instrument ([Lewbel, 2012](#)) built on the coupon. I set aside the quarters of the flight-to-quality episodes documented in the literature, from the 1987 crash to the 2022 tightening, when flight-to-safety demand shocks move yields directly: the regressions control for them, and the heteroskedasticity instrument is built only from calm-period variation. The first-stage F statistics are 7.7 for Treasuries, on two instruments whose overidentifying restriction is not rejected, and 18.1 for agency MBS, and I report a weak-instrument-robust Anderson–Rubin set next to every slope, describing a slope as different from zero only when its set excludes zero.

The estimated demand parameters show who absorbs supply in each market. In the Treasury market only two sector types have long-run slopes whose Anderson–Rubin sets exclude zero, the household residual and mutual funds; banks, insurers, pension funds, and foreign investors, who hold most of the float, do not respond. In the agency MBS market four of the eight do: the household residual, foreign investors, banks, and pension funds. Absorption is slow in both markets. A typical absorber completes only an eighth to a quarter of its one-year response in the quarter of the shock, pension funds being the one fast absorber, and several absorbers are still building a year later, the slow-moving capital of [Duffie \(2010\)](#) as a property of every sector that absorbs supply. In agency MBS the household residual and the arbitrageurs even trade the other way on impact, so that market has no marginal buyer in the quarter a supply shock lands.

The demand slopes and the composition of holdings together determine the slope of each market’s aggregate demand curve, which pins down the basis-point impact of a supply shift. In 2025Q1 the Treasury slope implies that removing 1 percent of the private float raises the 10-year yield by about 8 basis points, with a 95 percent band of 4 to 29, and the household

residual supplies more than four-fifths of the slope behind it. The agency MBS slope implies that removing 1 percent of the MBS float widens the mortgage spread by 2.0 basis points, with a band of 1.3 to 4.2, and unlike the Treasury slope it rests on many sector types rather than one. The Treasury slope has moved with that one sector’s presence: it fell by two-thirds in the early 2000s and stayed low through 2008, as foreign reserve managers accumulated Treasuries and the residual emptied, and it recovered after 2009 and again after 2022 as the float shifted back toward the residual and the hedge funds in it.

Lastly, I conduct a counterfactual analysis of Fed asset purchases using the estimated system. Equilibrium yields are jointly determined by the demand system and the market-clearing condition in the two markets the Fed trades, with the balance-sheet shocks measured as the Fed’s own transactions and bond supply responding to yields through net-issuance elasticities estimated from the same data. Across the four programs, QE1 through QE4, the Treasury impact runs from 8 to 12 basis points per 1 percent of the float removed, close to the 8 to 13 of [Eren et al. \(2026\)](#). QE1’s mortgage effect of 108 basis points sits at the top of the event-study record of 85 to 107. The 2017–2019 runoff raised Treasury yields by 23 basis points and mortgage yields by 31. The 2022–2025 runoff raised them by 65 and 76. Looking forward, a further \$1 trillion raises 10-year Treasury yields by 22 basis points and mortgage yields by 27; applied to \$27 trillion of marketable federal debt as it rolls over, the Treasury rise adds roughly \$60 billion a year to federal interest costs, and the mortgage rise about \$72 a month to the payment on a newly originated \$400,000 mortgage. The Treasury number is a band, not a point: it runs from 6 basis points if the runoff lands in a recession to 45 at the lower edge of the household slope’s confidence set.

This paper contributes to the literature on market macrostructure ([Haddad and Muir, 2025](#)), and in particular to demand-system asset pricing in bond markets.² [Koijen et al. \(2021\)](#) find that foreign investors, whose demand is most elastic, sell the largest share of bonds to the ECB during quantitative easing. [Jansen et al. \(2024\)](#) estimate sector-level

²Several papers estimate demand systems for bond markets without a direct focus on QE: [Fang et al. \(2025\)](#) for global sovereign debt and [Bretscher et al. \(2025\)](#) for corporate bonds.

Treasury demand within an equilibrium model of arbitrageurs and find that QE moves yields only insofar as investors perceive the purchases as persistent; my counterfactuals, which hold the balance-sheet change in place, correspond to their credible-commitment case, and their finding that hedge funds and dealers are the marginal Treasury investors is the high-frequency counterpart of the residual sector that absorbs Treasuries here. [Eren et al. \(2026\)](#) estimate demand for U.S. Treasuries, find that QE and QT have asymmetric yield effects, and extend the analysis to Japan and the U.K. [Fang and Xiao \(2025\)](#) extend the approach to three bond classes at the security level and find that for every unit of bonds the Fed purchases, mutual funds and banks purchase an additional 0.5 units, amplifying the effect of Fed policy; their granularity also recovers substitution my sector-level classes average over, such as between long-maturity Treasuries and agency MBS, close substitutes at matched duration. [Chaudhary et al. \(2026\)](#) decompose realized Treasury yield movements into investor-level drivers with a granular instrumental-variable estimator and find that aggregate Treasury demand is highly inelastic: a 1 percent shift in demand moves yields by 13 basis points, close to the upper end of the per-unit impacts estimated here. My main contribution is a detailed evaluation of QE and QT through a demand system. I price every program from QE1 to QT2, and a prospective runoff, in both markets the Fed trades; I separate the Treasury level from the mortgage spread, so that the composition of a program can be read off its effect on mortgage rates; and I trace how the yield effect builds and settles as slow capital arrives. Four features of the system make this possible. First, I measure demand from the Financial Accounts' transactions, complementing the market-value holdings used in prior work, so that the slopes of book-value holders such as banks and insurers reflect their trading. Second, I measure each sector's demand against its own outside asset, the margin the portfolio-balance channel operates on. Third, I let the yield slope differ by asset, and find that the set of active absorbers differs across markets. Fourth, I let demand adjust at a sector-specific speed, estimating the response horizon by horizon with local projections, which separates the impact and long-run effects of QE and QT.

This paper also contributes to the empirical literature evaluating the effects of QE on asset prices. Event studies find that announcements of the Fed’s purchase programs lowered 10-year Treasury yields by roughly 90 to 110 basis points around QE1 and mortgage rates by 40 to 100 basis points (Fuster and Willen, 2010; Gagnon et al., 2011; Hancock and Passmore, 2011; Krishnamurthy and Vissing-Jorgensen, 2011; Stroebel and Taylor, 2012; Krishnamurthy et al., 2013), although Christensen and Rudebusch (2012) attribute about 60 percent of the QE1 decline to lower expected short rates, so announcement effects mix the signaling and portfolio-balance channels. Time-series and term-structure studies map the Fed’s holdings into term premiums and find declines of roughly 30 to 45 basis points per program (D’Amico et al., 2012; Ihrig et al., 2018; Wei, 2022), and evidence from actual purchases, rather than announcements, points to persistent stock effects (D’Amico and King, 2013; Vissing-Jorgensen, 2021). These estimates, however, evaluate only the policies actually implemented, under the conditions prevailing at the time: they cannot speak to counterfactual policies, such as alternative purchase sizes or asset mixes, and are vulnerable to the Lucas critique, since reduced-form price responses need not be stable across policy regimes. I fill this gap by estimating structural demand parameters that are more stable across policy regimes and support counterfactual analysis for policies never implemented. Calibrated preferred-habitat models also support counterfactuals (Vayanos and Vila, 2021); relative to their calibration, which matches yield moments in a Treasury-only model, I estimate demand from observed sector transactions and cover both of the markets the Fed trades, so the counterfactuals inherit measured investor heterogeneity and the substitution between the two.

This paper also contributes to the literature on the transmission channels of balance sheet policy. QE transmits through signaling, portfolio rebalancing, and asset-specific channels such as safety, prepayment risk, and default risk (Krishnamurthy and Vissing-Jorgensen, 2011), as well as through bank lending (Chakraborty et al., 2020; Kandrach and Schlusche, 2021) and mortgage refinancing (Di Maggio et al., 2020). This paper focuses on portfolio

rebalancing: when the Fed purchases bonds, investors shift into imperfect substitutes. I quantify the channel by estimating each sector's substitution between the bonds the Fed trades and its outside asset. The estimates speak to the debate between a broad duration channel, under which purchases in any market move all long-term yields ([Gagnon et al., 2011](#)), and the narrow view that the impact concentrates in the purchased asset ([Krishnamurthy et al., 2013](#)). Each holds in part: Treasury purchases lower mortgage yields through the Treasury level, while the mortgage spread responds mainly to the MBS the Fed buys or runs off.

Finally, this paper contributes to the literature on the implementation and limits of QT. How far the central bank can shrink its balance sheet without side effects depends on both sides of that balance sheet. On the liability side, QT drains reserves from financial intermediaries and strains the repo market, so the demand for reserves and liquidity bounds the size of QT (e.g., [López-Salido and Vissing-Jorgensen, 2025](#); [Anbil et al., 2026](#); [Anderson et al., 2026](#); [Oguri and Pizzimenti, 2026](#)). This paper focuses instead on the asset side: as the Fed reduces its holdings of Treasuries and agency MBS, the remaining investors absorb the supply only at higher yields, and this yield effect, not reserve scarcity, limits QT. The effects of QT cannot simply be read off estimates of QE: on the liability side the asymmetry arises from reserve scarcity ([Acharya et al., 2024](#)), and on the asset side, in this paper, from the composition of the float, which shifts the set of investors that must absorb the supply, and from the state of the cycle in which the runoff lands. Assessing QT therefore requires knowing who holds bonds, which of them trade on yields, how steeply, and how fast their capital moves; the demand system estimated here provides exactly these inputs, and it says that for Treasuries the answer today rests on two sectors, one of which supplies four-fifths of the slope.

2 Data

The dataset covers the information required for the demand estimation: portfolio holdings and the transactions that change them, asset prices (yields), asset characteristics, and instrumental variables for asset prices.

2.1 Portfolio Holdings and Transactions

Sectors. I use the Federal Reserve’s Flow of Funds (the Financial Accounts of the United States, Z.1) to compute sector-level portfolio shares and to measure the transactions behind their changes. The Flow of Funds reports quarterly balance sheets for the U.S. financial sectors (e.g., life insurance companies, private pension funds). The sample runs from 1985Q1 to 2025Q1, a span that reaches back to the high-yield years of the late 1980s and, as Section 3.7 shows, is what identifies the marginal buyer of Treasuries. Over this span I work with 27 sectors: the 25 domestic sectors of the Financial Accounts and the rest of the world split into Foreign Official and Foreign Private holders. The split uses the Federal Reserve’s CSLT estimates of cross-border securities positions (Bertaut and Judson, 2014), which report monthly holdings and net transactions of U.S. Treasuries, agency bonds, corporate bonds, and equities by official and non-official foreign investors from 1984 onward. A separate hedge-fund sector from the Federal Reserve’s Enhanced Financial Accounts (EFA), built from SEC Form PF filings, is available only from 2013, so the sample keeps hedge funds within the household residual; Appendix D.6 uses the EFA to measure their share of it. Table 1 lists the sectors and their broader sector-type classifications (e.g., insurers, pension funds); the foreign official and private units both belong to the foreign type.

Assets. The two bond markets the Federal Reserve trades, Treasury securities and agency MBS, are the assets inside the demand system; the Flow-of-Funds asset classes are coarser than CUSIP-level data but detailed enough (e.g., Treasury securities, corporate equity, cash) to identify them. Together the sectors form a heterogeneous demand side, yet a few large

Table 1: **Sector Coverage and Outside Asset Definitions**

Type	Sector	Outside Asset
Banks	U.S.-Chartered Depositories	Loans
	Foreign Banking Offices	Total non-bond
	Credit Unions	Loans
	Banks in Affiliated Areas	Total non-bond
Insurers	Life Insurers	Corporate equities
	Property-Casualty Insurers	Corporate equities
Pension Funds	Private Pension Funds	Corporate equities
	State & Local Govt. Retirement Funds	Corporate equities
	Federal Govt. Retirement Funds	Total non-bond
Mutual Funds	Mutual Funds	Corporate equities
	Exchange-Traded Funds	Corporate equities
	Money Market Funds	Repo
	Closed-End Funds	Corporate equities
Arbitrageurs	Brokers and Dealers	Repo
	mREITs	Total non-bond
Foreign	Foreign Official	U.S. equities
	Foreign Private	U.S. equities
Household Residual	Household Residual & Nonprofits	Corporate equities
Other	Nonfinancial Business	Total non-bond
	Government-Sponsored Enterprises	Total non-bond
	Holding Companies	Total non-bond
	State & Local Governments	Total non-bond
	Federal Government	Total non-bond
	Agency/GSE Mortgage Pools	Total non-bond
	Finance Companies	Total non-bond
	ABS Issuers	Total non-bond
	Other Financial Business	Total non-bond

Notes: Sectors in the portfolio data and the outside asset against which each sector’s bond demand is measured, chosen as the non-bond asset the sector realistically trades off against bonds; “Total non-bond” is total financial assets net of all bond holdings, used where no single alternative dominates. The rest of the world is split into Foreign Official and Foreign Private holders with the Federal Reserve’s CSLT data. Hedge funds remain inside the household residual, because the Enhanced Financial Accounts begin only in 2013 and include offshore funds that the Financial Accounts assign to the rest of the world. Appendix D.1 explains each outside asset, and Appendix Table A.9 lists sources and table codes.

groups hold most of the bonds: throughout the sample, foreign investors and domestic financial institutions (banks, insurers, pension funds, and mutual funds) hold about 80% of the two markets, while arbitrageurs (e.g., dealers, hedge funds) hold only about 5%. Figure 1 reports each market’s ownership composition and total issuance over time, from 1985 to 2025. Because the ten-year yield prices coupon securities rather than bills, I measure the Treasury asset on the coupon market, using the non-bill holdings and transactions the Accounts publish for the six holders whose bills they break out (tables L.210 and F.210) and the total for the remaining sectors. Agency MBS is left as reported: table L.211 does not separate agency MBS from the debentures of the government-sponsored enterprises, about 17 percent of the total, which attenuates the estimated MBS slopes.

Transactions and Revaluation. The change in a sector’s portfolio mixes two things: the sector’s trading, and the revaluation of the positions it already holds when prices move. Demand estimation has to separate the second from the first. Eren et al. (2026) approximate the revaluation with a duration-based price change of the bond index. The Financial Accounts also record each sector’s revaluation of every instrument alongside its net purchases, so I can use the revaluation each sector reports directly. A recorded holding evolves as

$$H_{it}(n) = H_{i,t-1}(n) + F_{it}(n) + R_{it}(n), \quad (1)$$

where $H_{it}(n)$ is sector i ’s recorded holding of asset n , $F_{it}(n)$ its net purchases during quarter t , and $R_{it}(n)$ the revaluations and other volume changes. Taking logs, the growth of the holding splits exactly into a trading term and a revaluation term,

$$\ln \frac{H_{it}(n)}{H_{i,t-1}(n)} = \underbrace{\ln \left(1 + \frac{F_{it}(n)}{H_{i,t-1}(n)} \right)}_{g_{it}(n), \text{ trading}} + \underbrace{\ln \left(1 + \frac{R_{it}(n)}{H_{i,t-1}(n) + F_{it}(n)} \right)}_{r_{it}(n), \text{ revaluation}}, \quad (2)$$

and so does the change in the log share ratio against the outside asset:

$$\Delta \ln \frac{H_{it}(n)}{H_{it}(0)} = [g_{it}(n) - g_{it}(0)] + [r_{it}(n) - r_{it}(0)]. \quad (3)$$

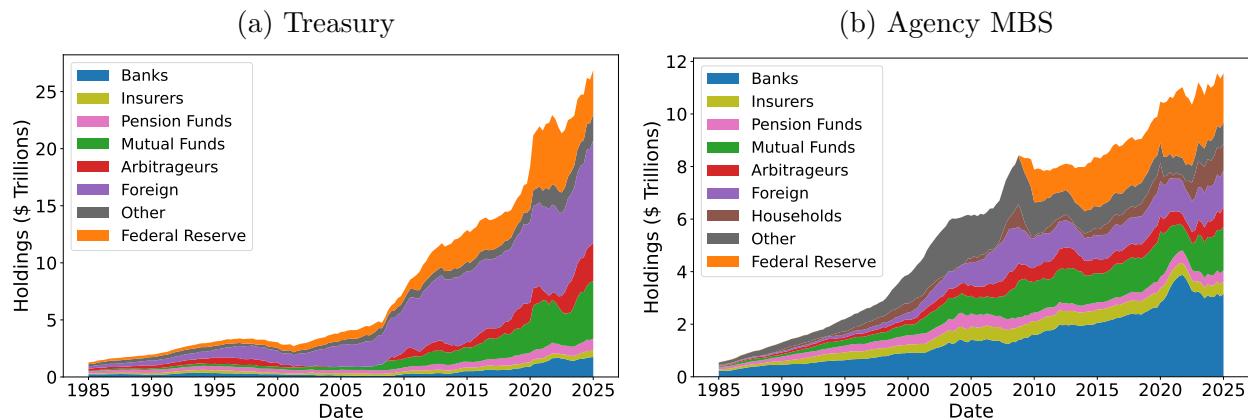
I remove the recorded revaluation and use the remainder as the dependent variable of the demand system:

$$\Delta q_{it}(n) \equiv \Delta \ln \frac{H_{it}(n)}{H_{it}(0)} - [r_{it}(n) - r_{it}(0)] = g_{it}(n) - g_{it}(0). \quad (4)$$

The two approaches coincide for a holder that marks its book to market, and the recorded flows add sector detail where accounting conventions differ: banks report held-to-maturity securities at amortized cost, insurers at statutory book value, and the rest of the world at market value, so the recorded pass-through of a price change into the level ranges from zero to nearly one across sector–asset cells (Appendix D.2). With the recorded revaluation removed, a sector that neither buys nor sells has $\Delta q_{it}(n) = 0$ whatever its accounting. I winsorize each cell’s transaction growth at its 1st and 99th percentiles, because a growth rate is uninformative on a near-zero base, and leave it undefined where the lagged position is below \$50 million.

Outside Asset. The outside asset is sector-specific: for each sector I designate the single non-bond class it trades off against bonds, and measure its bond demand relative to that class. Domestic banks, for example, rebalance between bonds and loans rather than between bonds and every other asset on their balance sheet, so loans are their outside asset. The last column of Table 1 reports each sector’s outside asset, and Appendix D.1 gives the rationale sector by sector. Appendix Figures A.6–A.6 show how each sector’s portfolio shares evolve over the sample.

Figure 1: **Ownership Structure of U.S. Bond Markets**



Note: This figure shows the ownership share of the two U.S. bond markets the Federal Reserve trades, Treasury securities and agency MBS, by institutional sector. Appendix A.1 describes the sector classification. In Panel (a) the Z.1 household residual in Treasury securities is reassigned to hedge funds within the Arbitrageurs sector, as these holdings predominantly reflect hedge funds engaged in the cash–futures basis trade. Panel (a) is the whole Treasury securities market, bills included, which is the market the Financial Accounts report; the demand system prices the coupon market, which excludes them.

Limitations. I acknowledge the limitations of the sector-level portfolio share data from the Flow of Funds. The data do not report within-class maturity variation, so I cannot estimate demand across precise duration buckets. Thus, this paper focuses on substitution across broad fixed-income asset classes rather than fine maturity choice.

2.2 Asset Characteristics

Yields. This paper uses one yield per asset class as its price. For Treasuries I use a representative constant-quality series that spans the full sample, the 10-year constant-maturity Treasury yield (DGS10) from FRED. A point yield at a representative maturity tracks the long end of the market, where habitat demand bites, more cleanly than a whole-market index does over a four-decade sample. For agency MBS I use an option-adjusted yield, the 10-year Treasury yield plus the option-adjusted spread of the Bloomberg agency-MBS index (available from 1990Q3, which sets the start of the MBS estimation sample): the index's own yield-to-worst embeds the prepayment-option premium, which moves with interest-rate

volatility rather than with the compensation investors require for holding MBS. Table [A.10](#) in the Appendix reports the sources and ticker codes. I aggregate all series to the quarterly frequency, taking the end-of-quarter value.

Other Characteristics. The system uses four asset characteristics: maturity, net issuance, duration, and coupon. The weighted time-to-maturity, modified duration, and weighted-average coupon of each index come from Bloomberg. Duration enters lagged one quarter, because its change is a near-mechanical function of the contemporaneous yield change ($dD \approx -D dy/(1 + y)$). The coupon is a slow-moving composition variable and is the exogenous shifter behind the Treasury heteroskedasticity instrument (Section [3.6](#)). Net issuance is measured from the Z.1 flow tables, as every holder’s transactions in the asset, the Federal Reserve included, over the lagged amount outstanding, rather than from the change in market-value outstanding, which is mostly revaluation and would enter the estimation as a function of the yield itself. Table [A.10](#) in the Appendix lists the sources for each asset class.

Macroeconomic Variables. I include a set of macroeconomic controls to absorb time-series variation in aggregate portfolio demand unrelated to asset prices. Specifically, I control for log real GDP, real GDP growth, core inflation, and the log of the U.S. dollar index, all obtained from the Federal Reserve Economic Data (FRED) database and aggregated to the quarterly frequency. I also include the one-quarter lag of log VIX as a measure of aggregate risk sentiment. Because VIX is jointly determined with asset prices during flight-to-safety episodes, I use the lagged value as a predetermined control to avoid simultaneity bias. The estimation also includes an indicator for the twelve crisis quarters in which the instruments are switched off (Section [3.6](#)).

2.3 Instruments

The main instruments come from two sources, as Section 3.6 explains: the unexpected part of the Fed’s asset purchases, for both markets the Fed trades; and a crisis-robust heteroskedasticity-based instrument built from the coupon characteristic already in the panel, which the Treasury block uses alongside it. A granular instrument built from the sectors’ holdings, maturity walls, and the monetary-policy surprises of the prior demand-system literature enter as diagnostics only. This subsection records the data sources.

Unexpected Federal Reserve Asset Purchases. The instrument for both markets is the unexpected part of the Fed’s asset purchases. Realized purchases are not a surprise, because once a program is announced they follow a published monthly schedule; the surprise is the revision to that schedule. For each of the twenty-one FOMC communications between November 2008 and May 2024 that changed the announced path of System Open Market Account purchases, I record the revision to expected net purchases of agency MBS and of Treasuries over the following four quarters, relative to the previously announced path, date it to the quarter in which the change first became public, and scale it by the amount outstanding. Appendix D.4 describes the construction, and Appendix Table A.16 lists every event.

Monetary Policy Announcement Shocks. High-frequency shocks identified from intraday market movements around FOMC announcements enter in two places. The LSAP factor of Swanson (2021) is a diagnostic only: it has no first stage in either market (Appendix Table A.18). The MBS-price surprise of Drukker (2025) identifies the agency MBS supply schedule of Section 5.2; the demand side uses neither.

2.4 Summary Statistics

Table 2 reports summary statistics for the merged dataset. Panel (a) presents average portfolio shares and assets under management (AUM) by sector type. Foreign investors allocate the largest share of their portfolios to Treasuries, 28.9 percent, and arbitrageurs and banks the largest to agency MBS, 11.5 and 10.8 percent. Households and pension funds hold small fixed-income shares despite their large aggregate AUM. Most of each market’s float sits with sectors that, Section 4 shows, do not trade on yields: in 2025Q1 foreign investors, banks, pension funds, and insurers together hold 56 percent of the private coupon float, and the two sectors whose active demand does respond to Treasury yields, the household residual and the mutual funds, hold about a seventh each. This concentration is what makes the slope of the Treasury market’s aggregate demand curve small. Panel (b) reports the characteristics of the two assets. Their yields average almost the same, 4.8 percent for Treasuries and 4.7 for agency MBS, but the Treasury index is the longer one, 8.1 years of maturity against 6.8 and 5.5 years of duration against 4.8, prepayment compressing the effective duration of MBS. The estimation sample covers 26 sectors observed quarterly from 1985Q1 to 2025Q1.

3 Empirical Framework

This section develops a demand system for the two U.S. bond markets the Federal Reserve trades, Treasury securities and agency MBS, identified from sector-level portfolio data. Four features set it apart from existing bond demand systems. First, demand is measured from transactions rather than from the change in the market value of holdings, so the yield coefficient is identified from trading. Second, the outside asset differs across sectors: I measure each sector’s bond demand against the non-bond class it actually rebalances toward, loans for banks, corporate equities for long-term investors, and repo for money funds and dealers (e.g., [Rodnyansky and Darmouni, 2017](#); [Kandrac and Schlusche, 2021](#); [Kurtzman et al., 2022](#)). Table 1 lists the outside asset for each sector and Appendix D.1 gives the

Table 2: **Summary Statistics**

(a): Portfolio Summary

Sector Type	Financial Assets (\$T)		Portfolio Weight (%)	
	Total	Bond	Treasury	Agency MBS
Banks	12.87	1.91	3.60	10.78
Insurers	6.24	0.59	4.42	7.08
Pension Funds	13.21	0.74	4.29	3.05
Mutual Funds	12.07	1.41	5.67	7.13
Arbitrageurs	3.22	0.45	3.39	11.50
Foreign	19.68	6.62	28.94	4.85
Household Residual	50.04	0.79	1.09	0.49
Other	30.91	1.52	2.23	3.72

(b): Asset Summary

Asset	Variable	Mean	Median	S.D.	Min	Max	N
Treasury	Yield (%)	4.77	4.44	2.39	0.66	11.65	161
	Maturity (years)	8.06	8.16	0.94	6.08	9.88	161
	Issuance (log \$M)	15.50	15.16	0.89	14.06	17.11	161
	Duration (years)	5.49	5.35	0.63	3.93	7.11	161
	Coupon (%)	5.29	4.97	2.83	1.57	11.55	161
Agency MBS	Yield (%)	4.71	4.54	2.03	1.31	9.60	139
	Maturity (years)	6.75	6.78	1.63	2.15	12.31	161
	Issuance (log \$M)	15.28	15.63	0.86	13.22	16.26	161
	Duration (years)	4.76	4.84	0.84	1.84	6.38	161
	Coupon (%)	6.06	5.50	2.32	2.60	10.73	161

Notes: Panel (a) reports sector-level financial assets and portfolio allocations for the two bond markets the system prices, drawn from the Z.1 Flow of Funds. The Treasury weight is the coupon market, the non-bill series of Section 2. Portfolio weights are time-series averages, weighted by sector financial assets. Panel (b) reports summary statistics for yield, maturity, issuance, duration, and coupon at the asset-quarter level, the unit at which the characteristics vary (N is the number of quarters; the agency-MBS yield begins in 1990Q3).

rationale. Third, each sector’s yield sensitivity differs by asset class, so the system can say who absorbs Treasuries and who absorbs agency MBS instead of imposing one slope per investor across both markets. Fourth, demand adjusts at a sector-specific speed, estimated horizon by horizon with local projections, which separates the impact slope from the long-run slope the counterfactual uses.

The sector-specific outside asset has a cost and a benefit. The cost is that the slopes are not directly comparable across sectors, since each is defined relative to a different outside asset; the aggregate demand slope of Section 3.4, the object the counterfactual requires, aggregates across sectors whatever outside asset each uses. The benefit is that the substitution margin is the one the portfolio-balance channel of QE and QT operates on. [Jansen et al. \(2024\)](#), [Fang et al. \(2025\)](#), and [Eren et al. \(2026\)](#) measure demand against other bonds, and investors substitute far more readily between one bond and another than between bonds and equities or loans, so a within-bond outside asset yields a more elastic demand than the bond-versus-non-bond margin that governs the price impact of central-bank purchases.

The use of aggregate shares is analogous to the market-level setup of [Berry et al. \(1995\)](#), without random coefficients, because all sectors face the same prices and characteristics at each date; Appendix A gives the mapping.

3.1 Setup

I index sector-level investors by $i = 1, \dots, I$; sectors include, for example, life insurers and mutual funds. Assets are indexed by $n = 0, \dots, N$, where $n = 0$ denotes the outside asset, which differs across sectors and comprises non-bond holdings such as equities and loans. The bond universe consists of Treasury securities and agency MBS, the two markets the Federal Reserve trades. Corporate and municipal bonds are not in it: no instrument for their yields survives the measurement discipline of Section 3.6, so their demand slopes cannot be estimated. They are not folded into the outside asset either: they sit outside the choice set. In the logit that leaves the estimates unchanged, because the log ratio of a

bond's share to the outside asset's share does not depend on which other alternatives the investor holds. Appendix D.1 uses them as a common outside asset instead. The yield of asset n in quarter t is $y_t(n)$. Each asset has a vector of observed characteristics $x_t(n)$: maturity, net issuance measured from transactions, and duration. The characteristics enter the estimation predetermined; in particular, duration enters lagged one quarter, because the change in modified duration is a near-mechanical function of the contemporaneous yield change (Section 2). The portfolio weight of investor i in asset n at time t is $w_{it}(n)$, with $\sum_{n=0}^N w_{it}(n) = 1$.

3.2 Portfolio Choice

I assume that portfolio shares follow a characteristics-based logit demand, under which the share is exponential in a linear index of characteristics (Kojen and Yogo, 2019).³ The portfolio weight $w_{it}(n)$ is

$$w_{it}(n) = \frac{\delta_{it}(n)}{1 + \sum_{n=1}^N \delta_{it}(n)}, \quad (5)$$

where

$$\delta_{it}(n) = \exp\left\{\beta_i(n) y_t(n) + \gamma_i' x_t(n) + \xi_{it}(n)\right\} \epsilon_{it}(n). \quad (6)$$

$\beta_i(n)$ is investor i 's demand semi-elasticity with respect to the yield of asset n , and γ_i (a vector) collects investor i 's loadings on asset characteristics. $\xi_{it}(n)$ denotes an unobserved demand shifter capturing asset-time variation not explained by observed characteristics, and $\epsilon_{it}(n)$ is the latent demand not explained by the observed yield or characteristics. The outside-asset weight is

$$w_{it}(0) = \frac{1}{1 + \sum_{n=1}^N \delta_{it}(n)}. \quad (7)$$

The yield slope carries an asset index. Existing bond demand systems impose $\beta_i(n) = \beta_i$: a one-percentage-point rise in the yield of any bond shifts investor i 's log-odds of holding

³Kojen and Yogo (2019) provide an asset pricing-based theoretical foundation for logit demand by deriving it from mean-variance portfolio choice under factor-structure assumptions.

it by the same amount. I let the slope differ by asset, the alternative-specific-coefficient logit familiar from discrete choice. The share structure is unchanged: substitution across assets still follows the logit’s proportional pattern, and the outside asset still absorbs the balance, so the own- and cross-yield responses of Section 3.4 keep their closed forms. What the generalization drops is only the equal-slope restriction, and Section 4 shows that the data reject it: mutual funds, for example, absorb Treasuries but not the mortgage spread, and banks absorb agency MBS but not Treasuries. The common-slope system is the special case $\beta_i(n) = \beta_i$.

The price that enters each block, $y_t(n)$, is the one that market itself sets: the ten-year yield for Treasuries and, for agency MBS, the option-adjusted spread over it, since the MBS yield is the Treasury yield plus that spread and the Treasury leg carries eight times its variance. Writing MBS demand on the yield level instead would impose that a sector’s share responds equally to the Treasury yield and to the spread; Section 4 tests the two restrictions against each other. With this choice the clearing system of Section 5 is block-recursive: the Treasury market sets the level, the agency MBS market sets the spread, and the mortgage-relevant yield is their sum.

Combining (5) and (7) yields the estimable specification:

$$\begin{aligned} \ln \frac{w_{it}(n)}{w_{it}(0)} &= \ln \delta_{it}(n) \\ &= \beta_i(n) y_t(n) + \gamma'_i x_t(n) + \xi_{it}(n) + \varepsilon_{it}(n). \end{aligned} \quad (8)$$

The left-hand side is the log portfolio share of asset n relative to the outside asset. The right-hand side is a linear combination of the asset’s yield $y_t(n)$, characteristics $x_t(n)$, and asset–time demand shifter $\xi_{it}(n)$ with estimable coefficients; $\varepsilon_{it}(n)$ is an idiosyncratic demand shock, the log of the latent demand $\epsilon_{it}(n)$. I parameterize the asset–time demand shifter as

$$\xi_{it}(n) = \alpha_n + \nu'_{in} W_t + \tilde{\xi}_{it}(n), \quad (9)$$

where W_t is a vector of macroeconomic controls, ν_{in} captures sector- and asset-specific loadings on macro conditions, and $\tilde{\xi}_{it}(n)$ is the residual unobserved demand shifter. I follow [Jansen et al. \(2024\)](#) and [Eren et al. \(2026\)](#) in parameterizing the asset-time demand shifter via macro controls rather than asset-time fixed effects, which would absorb the time-series variation in the instruments.

I estimate the demand equation in first differences. Differencing (8) is an algebraic transformation that leaves every slope unchanged,

$$\Delta \ln \frac{w_{it}(n)}{w_{it}(0)} = \beta_i(n) \Delta y_t(n) + \gamma'_i \Delta x_t(n) + \nu'_{in} \Delta W_t + \Delta \varepsilon_{it}(n). \quad (10)$$

In estimation the left-hand side is measured net of the revaluation of positions already held, by $\Delta q_{it}(n)$ of Equation (4). Differencing has two benefits. First, it removes the asset fixed effect α_n . Second, it makes both sides of the equation stationary.

3.3 Rebalancing Speed

Equation (10) recovers the semi-elasticity $\beta_i(n)$ only if sectors complete their rebalancing within the quarter, and they do not: asset-liability mandates, illiquid positions, and the slow-moving capital of [Duffie \(2010\)](#) spread the response over several quarters. Estimated on one-quarter changes, (10) delivers only the *impact* response. I therefore treat (8) as the portfolio a sector ultimately wants and let observed positions converge to it at a sector-specific speed, as [van der Beck \(2026\)](#) do for equity institutions, and trace the adjustment with instrumental-variables local projections ([Jordà, 2005](#)):

$$\Delta^{(h)} \ln \frac{w_{i,t+h}(n)}{w_{i,t+h}(0)} = \beta_i^{(h)}(n) \Delta y_t(n) + \gamma'_i \Delta x_t(n) + \nu'_{in} \Delta W_t + \tau_{in} t + u_{it}^{(h)}(n), \quad (11)$$

where $\Delta^{(h)}$ takes the change from quarter $t-1$ to quarter $t+h$, measured as the cumulative transaction growth $\sum_{s=t}^{t+h} \Delta q_{is}(n)$; $\Delta y_t(n)$ is instrumented as in Section 3.6; the linear trend $\tau_{in} t$ absorbs the secular comovement of the long sample; and standard errors are Newey–

West with bandwidth h (Newey and West, 1987). Only the response window extends with h ; the yield change stays a one-quarter change, so the specification does not reintroduce the persistence that differencing removes. At $h = 0$, (11) is (10) and $\beta_i^{(0)}(n)$ is the impact slope; as h grows, $\beta_i^{(h)}(n)$ rises toward the full response. I estimate (11) for each sector type and asset, pooling the member sectors of the type.

I read the long-run slope one year out, at $h = 4$, a horizon fixed in advance, and take it as the semi-elasticity $\beta_i(n)$ of (8): whatever the frictions, once rebalancing has run its course the allocation is the one the static system describes. For absorbers that are still adjusting after a year, the one-year slope is a lower bound on that response. The speed of adjustment is a friction, not a demand parameter, so the aggregate demand slope and the counterfactual equilibrium use $\beta_i(n)$, and the transition to that equilibrium uses the path $\beta_i^{(h)}(n)$.

3.4 Demand Elasticity and the Aggregate Demand Slope

The logit demand system yields a closed-form expression for the yield elasticity of demand. The elasticity of the portfolio share of asset n with respect to the yield of asset m is

$$\eta_{it}(m, n) = \frac{\partial \log w_{it}(n)}{\partial \log y_t(m)} = \begin{cases} \beta_i(n) \cdot (1 - w_{it}(n)) \cdot y_t(n) & m = n \\ -\beta_i(m) \cdot w_{it}(m) \cdot y_t(m) & m \neq n, \end{cases} \quad (12)$$

where $\beta_i(n)$ is the long-run yield sensitivity of sector i for asset n from Section 3.3, $w_{it}(n)$ is the portfolio share of asset n , and $y_t(n)$ is the yield of asset n in quarter t . The own-elasticity ($m = n$) gives the percentage change in the portfolio share per one-percent rise in the asset's own yield; the cross-elasticity ($m \neq n$) is the spillover to other assets, which scales with the shocked asset's slope and portfolio share.

The object that governs the yield effect of a supply shift is not the elasticity but a yield-free slope. Linearizing the market-clearing condition of the next subsection around the observed equilibrium, a supply shift $\Delta S_t(n)$ must be absorbed by the sectors' share

adjustments, $\sum_i A_{it} \beta_i(n) w_{it}(n) (1 - w_{it}(n)) \Delta y_t(n) = \Delta S_t(n)$, so removing 1 percent of asset n 's private float raises its yield by

$$\Delta y_t(n) = \frac{1}{\sum_i s_{it}(n) \beta_i(n) (1 - w_{it}(n))} \text{ basis points}, \quad (13)$$

where $s_{it}(n) = A_{it} w_{it}(n) / S_t(n)$ is sector i 's share of the float and A_{it} its assets under management. The denominator is the slope of the aggregate demand curve for market n , which I call the aggregate demand slope. It aggregates the holders' own-asset sensitivities weighted by their share of the float: a market whose float sits with sectors that do not trade on yields has a small slope and a large price impact, whatever those sectors' behavior in other markets. Investors respond to yields point for point, so the yield level cancels and the basis-point impact is pinned down by this slope alone; its movement over time reflects the composition of holdings. Equation (13) is the first-order version of the counterfactual in Section 5, which clears the full nonlinear system.

3.5 Market Clearing

To close the asset pricing model, the market for each asset n clears in each period t :

$$\sum_i A_{it} \cdot w_{it}(n) = S_t(n), \quad (14)$$

where A_{it} is investor i 's total wealth at time t and $S_t(n)$ denotes the outstanding supply of asset n at time t . Market clearing holds in this study because, by construction, the aggregate share data cover all investors and sectors. The counterfactual imposes it in both markets.

3.6 Identification

The yield is endogenous: sector-level demand shocks move prices, the more so in a setting that aggregates investors into sectors rather than assuming atomistic behavior. I instrument

each asset's yield with an instrument specific to that asset, and require that its variation move portfolio shares only through that yield, after macroeconomic controls. Because the estimating equations are the differenced specification (10) and the local projections (11), whose horizon- h error cumulates the latent-demand revisions from t through $t + h$, the restriction applies over the projection window:

$$\mathbb{E}[\Delta \varepsilon_{i,t+s}(n) \mid Z_t(n)] = 0, \quad s = 0, 1, \dots, h. \quad (15)$$

At $h = 0$ this is the standard contemporaneous exclusion restriction; at longer horizons it adds the lead-exogeneity condition of LP-IV (Stock and Watson, 2018), that the instrument be unpredictable not only for current latent demand but for its revisions over the year the response cumulates.

Treasury Yields. No external supply shifter is available for Treasury yields over four decades: the Fed's own transactions track the yield cycle, maturity walls (Almeida et al., 2012; Harford et al., 2014) are predetermined but too small to be relevant, realized issuance responds to rates, and foreign demand chases yields. I therefore identify the Treasury slopes with the heteroskedasticity-based instrument of Lewbel (2012). At the asset-quarter level I regress the yield change on the differenced characteristics, take the residual $\hat{e}_t(n)$, and interact it with the demeaned coupon,

$$Z_t(n) = [\Delta x_t(n) - \overline{\Delta x(n)}] \hat{e}_t(n). \quad (16)$$

The instrument is valid if the characteristic is uncorrelated with the product of the demand and yield shocks, and relevant if the variance of the yield shock moves with the characteristic. The two conditions pull against each other in bond markets: yield-shock variance is highest in crisis quarters, and in exactly those quarters flight to quality makes demand and yield shocks covary. I therefore identify from calm-period heteroskedasticity alone. The crisis quarters are

those of the flight-to-quality episodes documented in the literature, the 1987 crash (1987Q4), the LTCM collapse (1998Q3), September 11 (2001Q3), the 2008 panic (2008Q3–Q4), the U.S. downgrade and euro-area crisis (2011Q3), the March 2020 dash for cash (2020Q1–Q2), and the 2022 tightening (2022Q1–Q4); I set the instrument to zero in these twelve quarters and control for them in the second stage. Alongside it I use the Treasury leg of the purchase-path revisions described next, dated to the quarter in which each change first became public rather than to the statement that formalizes it, since the Committee signals a change before it votes on it. Neither instrument is strong, so a weak-instrument-robust Anderson–Rubin set ([Anderson and Rubin, 1949](#); [Andrews et al., 2019](#)) accompanies every slope.

Agency MBS Yields. The Fed is the one buyer of agency MBS whose purchases are not a response to any private sector’s demand, but its *realized* purchases follow a monthly schedule announced in advance and carry no surprise. The information is in the revisions to that schedule. For each of the twenty-one FOMC communications between November 2008 and May 2024 that changed the announced path of System Open Market Account purchases, I record the implied revision to expected net purchases over the following four quarters, separately for agency MBS and for Treasuries and scaled by the outstanding stock. An asset-specific supply shifter has to move the price of its own market and leave the other’s untouched, and [Section 4](#) reports that placebo, along with the same test applied to the alternatives. The instrument is set to zero in the same twelve crisis quarters as the heteroskedasticity instrument, where the announcement and the flight to quality are inseparable. [Appendix D.4](#) describes the construction and lists the events, and [Appendix D.5](#) reports the menu of alternatives and the checks that the revisions do not respond to the previous quarter’s yields or macroeconomic conditions.

3.7 Non-Logit Demand Sectors

The demand system asks each sector to choose between a bond and an outside asset at the bond's yield. Two sectors do not fit that description, and both of them hold Treasuries.

Money market funds hold bills. Rule 2a-7 caps a money fund's weighted average maturity at sixty days and each holding at 397 days, so the ten-year yield is not the price its Treasury demand responds to. Bills are 78 percent of money funds' Treasury holdings today and 51 percent on average since 1974, against 0.7 percent for the long-duration mutual funds with which the Financial Accounts group them. I therefore estimate the mutual-fund type without money funds and treat the money-fund cell as inelastic in the clearing: their holdings stay in the float and their response is zero.

Dealers are intermediaries. A dealer's Treasury book is inventory financed by repo, not a portfolio choice: when yields rise the book loses value and a levered intermediary sells to restore its balance sheet. An intermediary holding leverage constant would sell

$$\frac{\partial \ln H_t(\text{dealer}, \text{Treasury})}{\partial y_t} = -\lambda_t D_t s_t^{\text{bond}}/100 =: \text{rule}_t, \quad (17)$$

where λ_t is leverage from the Financial Accounts, D_t the index duration, and s_t^{bond} the bond share of financial assets, all lagged one quarter so that rule_t is predetermined. I replace the dealer cell's free slope with φrule_t and estimate the single parameter φ on the same instruments, using $Z_t \text{rule}_t$ as the instrument for $\text{rule}_t \Delta y_t$, so the cell's contribution moves with the observed balance sheet. The rule is applied to the Treasury cell only; Section 4 reports φ in both markets.

The estimation sample is part of the identification. The household residual's Treasury slope is estimated on the 1985–2025 span, because the 2010–2025 window alone does not identify it (Section 4 and Appendix D.3), so the Treasury counterfactuals rest on the pre-2010 behavior of that sector. The agency MBS absorbers are identified in both windows.

3.8 Scope of the Demand-System Approach

The answers here are valid within the known limits of the demand-system approach (Fuchs et al., 2023; Kojen and Yogo, 2025). First, a logit demand system need not correspond to a valid stochastic discount factor or rule out arbitrage (Fuchs et al., 2023; Kocherlakota, 2025); I use it as a description of how holders trade, not as a model of prices derived from preferences. Second, the logit restricts substitution: letting the slope differ by asset relaxes the restriction within each holder, but the cross-asset responses still follow from the portfolio shares rather than being estimated separately, and the counterfactuals inherit that assumption. Third, identifying a demand system requires as many independent sources of price variation as priced assets (Fuchs et al., 2025). With two asset classes and an instrument set for each, the system meets that requirement, and because the instruments are aggregate time-series shocks, the slopes carry the aggregate response of demand rather than only its cross-section (Haddad et al., 2025). Finally, the slopes are identified from quarterly variation around the observed equilibrium, while the largest programs move supply far outside it, so the counterfactuals are most reliable per unit of supply and least reliable for the largest episodes.

4 Results

4.1 First Stage

Table 3 reports the first stage on the 1985–2025 asset–quarter panel, with the same controls as the second stage and Newey–West standard errors. Both markets the Fed trades are identified by two instruments each of which has the sign theory predicts. The Treasury block pairs the coupon interaction ($t = -3.7$) with the revision to the Fed’s announced Treasury purchase path ($t = -1.8$) for a joint F of 7.7, and the overidentifying restriction survives in every sector type. The agency MBS block is the stronger of the two: an upward

revision to the announced MBS purchase path compresses the option-adjusted spread with an F of 18.1 ($t = -4.3$). The Fed’s realized transactions reach only 3.1, because the realized flow follows the schedule the announcement already set. The revision passes the cross-asset placebo an asset-specific supply shifter has to pass, leaving the Treasury yield at an F of 1.0, and every alternative shifter I examined fails it (Appendix Table A.18). Because the yield varies only at the asset–quarter level, the first stage is common to every sector, so differences in the estimated slopes reflect demand heterogeneity rather than differences in the identifying variation. I read every slope through its weak-instrument-robust Anderson–Rubin set and call a slope different from zero only when that set excludes zero. Any remaining bias has a known direction: weak instruments pull two-stage least squares toward ordinary least squares, which the simultaneity of demand pushes down, so the point estimates are if anything lower bounds.

Two measurement choices carry the identification. For Treasuries the market-value granular instrument reaches a first stage of 10.4 and looks like the strongest instrument available; built from the donors’ transactions it reaches 4.3, so most of its relevance was the revaluation that also violates its exclusion restriction, and I drop it. For agency MBS, replacing the change in the market value of the Fed’s holdings with its transactions lowers the first stage from 7.3 to 3.5, and replacing the realized transactions with the revisions to the announced purchase path raises it to 18.1: the realized flow is anticipated and the revisions are not.

Three further checks bear on the identification (Appendix Table A.17). First, a three-donor Treasury block leaves the long-run slopes essentially unchanged at a first-stage F of 6.4, so the donor choice affects precision rather than the point estimates. Second, adding the Fed’s realized transactions to the purchase-path revision raises the agency MBS first stage from 18.1 to 22.1, but the overidentification test then rejects in four of the eight sector types, banks, insurers, mutual funds, and arbitrageurs, at $p \leq 0.05$: the anticipated flow and the announcement surprise disagree about what the data say. I keep the single instrument, just-identified, and read its slopes through their Anderson–Rubin sets. Third, LIML and

Table 3: **First Stage**

Asset	Instrument	Coefficient	First-stage F	N
Treasury	Lewbel (coupon)	-4.346^{***} (-3.73)	7.7	159
	Fed Treasury purchase-path revision	-1.750^* (-1.82)		
Agency MBS	Fed MBS purchase-path revision	-5.746^{***} (-4.25)	18.1	138
Joint F (all instruments)			8.0	297

Notes: This table shows the first stage, estimated asset by asset on 1985Q1–2025Q1 quarterly data, with Newey–West t -statistics (Bartlett kernel, four lags) in parentheses; the yield varies only at the asset–quarter level, so the first stage is common across sectors. Treasury yields are instrumented by a crisis-robust [Lewbel \(2012\)](#) heteroskedasticity instrument, the demeaned change in the coupon interacted with the first-step residual and set to zero in the twelve crisis quarters of Section 3.6 (1987Q4, 1998Q3, 2001Q3, 2008Q3–Q4, 2011Q3, 2020Q1–Q2, 2022Q1–Q4), together with the revision to the Fed’s announced four-quarter Treasury purchase path, scaled by the outstanding stock and zeroed in the same quarters; the agency MBS option-adjusted spread (available from 1990Q3, hence $N = 138$) by the revision to the Fed’s announced four-quarter purchase path, scaled by the outstanding stock and zeroed in the same twelve quarters. All regressions include the second-stage controls: the first-differenced asset characteristics (maturity, transaction-based net issuance, lagged duration), the differenced macro controls, an asset-specific trend, and the crisis indicator. The first-stage F is the Newey–West Wald statistic of each asset’s own instruments jointly. The final row reports the joint F of all instruments on the stacked asset $\times$ quarter panel with asset fixed effects; because the instrument matrix is block-diagonal, this statistic averages the blocks, and the per-asset F is the informative measure of each yield’s identification. Anderson–Rubin sets accompany every slope in Table 4.

Fuller estimates coincide with two-stage least squares in every cell, so the weak-instrument bias of the point estimates is not the issue; the width of the sets is.

4.2 Demand Slopes by Sector

Table 4 reports, for each sector type and each of the two markets the Fed trades, the impact slope $\beta_i^{(0)}(n)$, the long-run slope $\beta_i^{(4)}(n)$, and the 95 percent Anderson–Rubin set for the long-run slope. I take the long-run slope as the cumulative response one year after the shock, a horizon fixed in advance by which most of each absorber’s response has arrived; Figure 2 shows the paths to two years. Each market has its own, thin set of active absorbers, and the set differs across markets.

In the Treasury market the marginal buyer is the household residual, whose long-run slope is 0.73: a one-percentage-point rise in the 10-year yield raises its Treasury position by about one percent relative to its equity holdings. Mutual funds are second at 0.28, once money market funds are taken out of that sector type (Section 3.7); the money-fund cell has a slope of -0.56 against the ten-year yield, and pooling it in removes the type’s slope, which falls to 0.02. Banks are third at 0.09, and the rest are flat: pension funds at 0.03, insurers at -0.04 , and foreign investors, the largest holders, at zero. Only the first two Anderson–Rubin sets exclude zero. The arbitrageur cell, dealers and mortgage REITs, carries a large negative slope, -1.67 , which is the momentum of an inventory and basis book rather than a demand curve; the balance-sheet rule of Equation (17) replaces it, with $\varphi = 4.12$ (standard error 2.53), procyclical leverage in the sense of Adrian and Shin (2010), implying a slope of -0.71 at the 2025Q1 balance sheet. On agency MBS the same regression returns $\varphi = -4.16$, the opposite of deleveraging, so the rule is applied to the Treasury cell only. The foreign type pools two units that offset each other: reserve managers are flat, at 0.06, and private foreign investors chase returns, at -0.06 .

The household residual is what the Financial Accounts assign to no measured sector, and after 2010 that includes the hedge funds whose cash–futures basis positions dominate

its Treasury flows (Jansen et al., 2024). The Enhanced Financial Accounts show how far this has gone: hedge funds hold 122 percent of the residual’s coupon Treasuries on average over 2013–2025, and the residual’s Treasury transactions load on the quarterly change in hedge-fund holdings with a coefficient of 0.62 (Appendix D.6). The two are the same position, which matters because the slope is identified where the residual was households: the 2010–2025 window on its own gives -0.84 with a standard error of 0.82, an interval that spans every value the counterfactual could use. Whether a levered arbitrageur absorbs supply the way a buy-and-hold household did is the open question behind the Treasury band of Section 5. The cell is also the one most exposed to the measurement of transaction growth, because the residual itself is near zero or negative between 2004 and 2008 (Appendix D.2).

In the agency MBS market, where the price is the spread, every sector type slopes the right way. The household residual leads at 2.33, followed by foreign investors at 0.50, banks at 0.48, and pension funds at 0.26; mutual funds, the arbitrageurs, and insurers are positive but imprecise. Four of the eight sets exclude zero, against two in the Treasury market. That every slope is positive is not imposed: with the MBS yield level as the price, banks, pension funds, and the residual sectors carry negative long-run slopes, which is what a demand curve should never do. The data also settle the choice of price directly. Entering the Treasury yield and the spread separately, each with its own instrument, the level coefficient is not different from zero at the 5 percent level in any sector type, while the equality a yield specification imposes is rejected for pension funds ($p = 0.03$) and foreign investors ($p = 0.05$). Foreign absorption runs through both units, the official sector at 0.68 and the private sector at 0.32. In the 2010–2025 window the household residual, the arbitrageurs, banks, and pension funds keep their signs, while mutual funds turn negative and insurers and foreign investors sit at zero.

The data also reject the equal-slope restriction of the standard system, at the 10 percent level. A minimum-distance test of the restriction that each type’s Treasury and agency MBS slopes are equal gives 15.1 on 8 degrees of freedom ($p = 0.06$), driven by foreign investors,

pension funds, and the household residual (Appendix Table A.22). An intermediate structure, $\beta_i(n) = \beta_i \theta(n)$, is not rejected ($p = 0.29$), with loadings of 0.32 on the Treasury yield against 1.38 on the agency MBS spread: the same investor responds four times as strongly to a basis point of spread as to a basis point of level. The counterfactual uses the unrestricted slopes.

4.3 Rebalancing Speed

Figure 2 traces the horizon response $\beta_i^{(h)}(n)$ of every sector type in both markets, two years out. Every absorber builds its position slowly, and in the agency MBS market the build is monotone quarter by quarter. Banks move from 0.12 on impact to 0.25, 0.41, and 0.48 at one year; foreign investors from 0.09 to 0.24, 0.35, and 0.50; insurers from 0.02 to 0.04. The Treasury absorbers do the same, the household residual moving from 0.10 on impact to 0.73 at one year and mutual funds from 0.07 to 0.28. For these cells the impact response is between an eighth and a quarter of the one-year response, and pension funds are the one fast absorber, at 0.16 of 0.26 within the quarter. The slow-moving capital of Duffie (2010) is thus a property of every sector that absorbs supply, not of a particular investor class, and it operates at the scale of quarters. Several absorbers keep building beyond the first year: banks' and foreign investors' agency MBS slopes and the mutual funds' Treasury slope rise by a third to two-thirds more by $h = 8$, although the estimates there are less precise. The one-year slope the counterfactual uses therefore understates their full response, so if anything the counterfactual overstates the long-run yield effects.

Two cells trade the other way in the quarter of the shock and reverse afterwards. The household residual's agency MBS slope is -1.04 on impact and $+2.33$ at one year, and the arbitrageurs' is -0.11 and $+0.15$; both are momentum in the shock quarter and rebalancing after it. Their weight is large enough to matter for the market: at the 2025Q1 composition the agency MBS aggregate demand slope is slightly *negative* on impact and turns positive from the first quarter on. The MBS market has no marginal buyer in the quarter a supply

Table 4: Demand Slopes by Sector Type and Asset

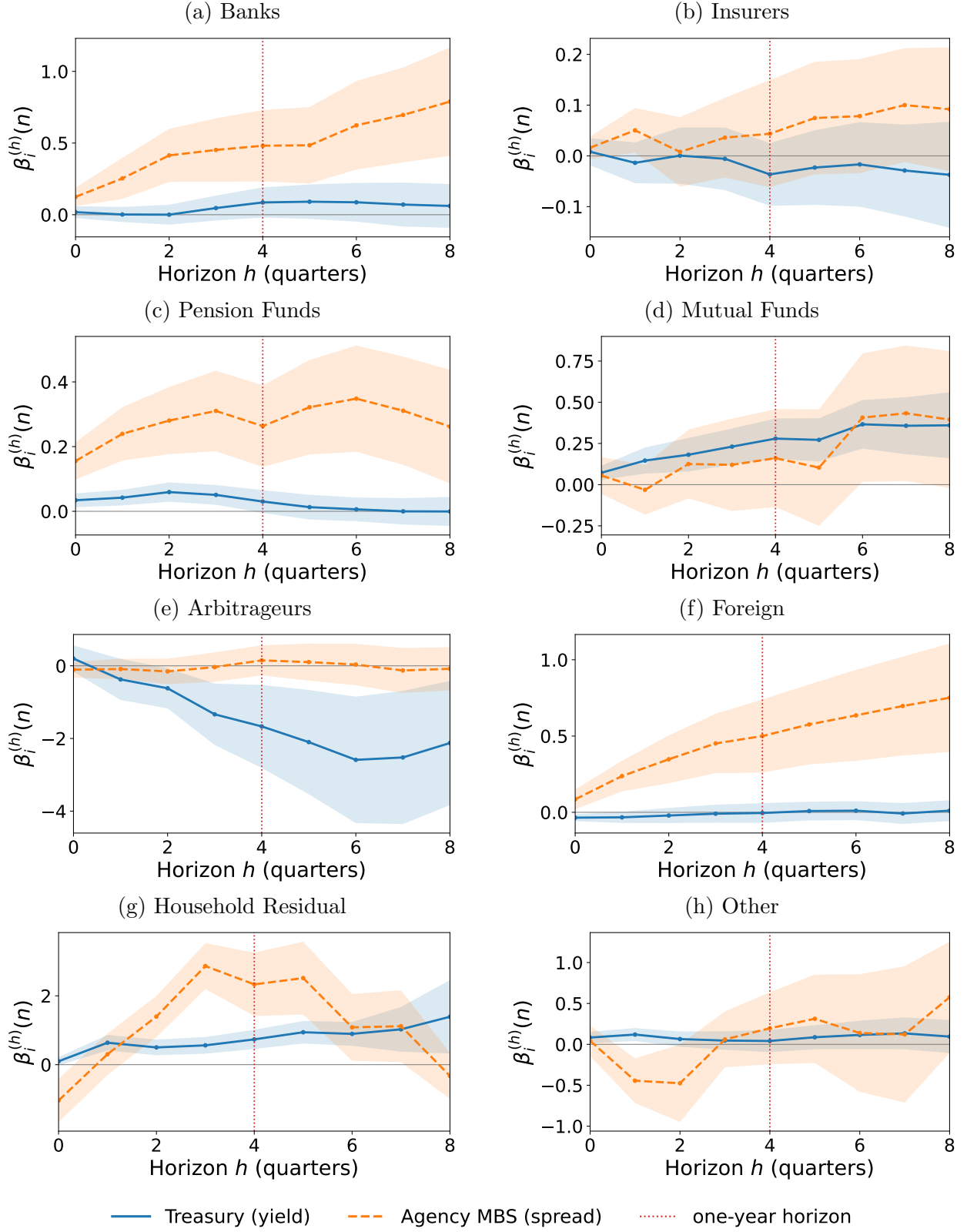
Sector type	Treasury (yield)		Agency MBS (spread)	
	$\beta^{(0)}$	$\beta^{(4)}$	$\beta^{(0)}$	$\beta^{(4)}$
Banks	+0.02 (0.04) [-0.06, 0.36]	+0.09 (0.11)	+0.12 (0.06) [0.03, 0.98]	+0.48 (0.25)
Insurers	+0.01 (0.03) [-0.16, 0.14]	-0.04 (0.06)	+0.02 (0.02) [-0.18, 0.24]	+0.04 (0.11)
Pension Funds	+0.03 (0.02) [-0.04, 0.11]	+0.03 (0.04)	+0.16 (0.06) [0.04, 0.54]	+0.26 (0.13)
Mutual Funds	+0.07 (0.04) [0.05, 0.65]	+0.28 (0.12)	+0.06 (0.11) [-0.37, 0.81]	+0.16 (0.30)
Arbitrageurs	+0.20 (0.36) [-8.52, 0.16]	-1.67 (1.14)	-0.11 (0.22) [-0.76, 0.90]	+0.15 (0.41)
Foreign	-0.04 (0.02) [-0.10, 0.12]	-0.00 (0.06)	+0.09 (0.06) [0.12, 1.12]	+0.50 (0.24)
Household Residual	+0.10 (0.12) [0.18, 1.91]	+0.73 (0.28)	-1.04 (0.63) [1.04, 5.10]	+2.33 (0.92)
Other	+0.08 (0.07) [-0.26, 0.34]	+0.04 (0.14)	+0.04 (0.19) [-0.68, 1.08]	+0.20 (0.44)
N	3,259		2,864	

Notes: Local projections of Equation (11), estimated separately for each sector type and asset on 1985Q1–2025Q1 (agency MBS from 1990Q3), pooling the type’s member sectors, with the yield instrumented as in Table 3. $\beta^{(0)}$ is the impact slope and $\beta^{(4)}$ the cumulative response one year after the shock, with Newey–West standard errors in parentheses; the bracketed row is the 95 percent Anderson–Rubin set for $\beta^{(4)}$, u marking a set that reaches the edge of the grid and $\emptyset$ an empty set. N is the number of sector–quarter observations. The two columns are not in the same units: a Treasury slope is the response to a percentage point of yield, an agency MBS slope the response to a percentage point of spread. The counterfactual uses $\beta^{(4)}$ except in the two cells of Section 3.7, where money market funds enter with a zero slope and the broker-dealer Treasury book follows the leverage rule. Since the mortgage REITs in that type hold no Treasuries, the arbitrageur Treasury slope reported here is never used.

shock lands, and acquires one a quarter later. Section 5 reports the consequence, which is that the impact-quarter equilibrium of a supply shift is not defined and the yield path of a runoff becomes defined only from the following quarter.

A static quarterly regression would therefore understate the absorbers' demand four- to eightfold. The distinction matters for the counterfactual: the impact response governs the initial dislocation, the long-run slope where yields settle, and the gap between them is the overshoot a balance-sheet shock produces before slow capital arrives.

Figure 2: **Rebalancing Paths by Sector Type**



Note: Horizon responses $\beta_i^{(h)}(n)$ from Equation (11), 1985Q1–2025Q1 (agency MBS from 1990Q3), with ± 1 Newey–West standard error bands; the dotted line marks $h = 4$, where the counterfactual reads the long-run slope. The yield and the spread differ in units, so lines within a panel compare in shape, not level.

4.4 The Aggregate Demand Slope

The slopes and the composition of holdings together determine how much a market’s yield moves per unit of supply. Figure 3 plots the implied price impact of Equation (13), the inverse of the aggregate demand slope $\sum_i s_{it}(n) \beta_i(n) (1 - w_{it}(n))$, for Treasuries and agency MBS from 1985 to 2025, and Table 5 reports the 2025Q1 slopes and impacts.

The Treasury slope is thin and has been thin since the turn of the century. In 2025Q1 it is 0.130 (standard error 0.049, $t = 2.7$), so removing 1 percent of the private float raises the 10-year yield by about 8 basis points, with a Fieller band of [4.4, 28.9] that is finite at both ends. The composition explains the number. Foreign investors hold 40 percent of the float and do not respond. Banks, pension funds, and insurers together hold 16 percent and contribute 6 percent of the slope. The household residual holds 15 percent and, with its slope of 0.73, supplies 82 percent of it; the mutual funds hold 14 percent and supply the remaining 22 percent; the arbitrageurs subtract 13 percent. The market’s marginal buyers are two sectors holding less than a third of the float between them. Over time the slope tracks that sector’s presence: it averaged 0.12 from 1985 to 1999 and fell to 0.042 between 2005 and 2008, reaching its low of 0.022 in 2005Q1, as foreign official reserve managers accumulated Treasuries and the residual emptied. It averaged 0.07 over the QE decade, touching 0.04 in 2014, and has averaged 0.12 since the float shifted back toward the residual, and the hedge funds inside it, after 2022. In the years when the slope is near zero the static system has almost no marginal buyer, and the yield effect of a given supply shift is correspondingly large; Section 5 shows what that means for the 2012–2014 and 2017–2019 episodes.

The agency MBS aggregate demand slope is four times larger and, unlike the Treasury slope, it is estimated with precision. In 2025Q1 it is 0.512 (standard error 0.140, $t = 3.7$), so removing 1 percent of the MBS float widens the option-adjusted spread by 2.0 basis points, with a Fieller band of [1.3, 4.2] that is finite at both ends. The precision comes from the spread specification: under the yield specification the same calculation gives 0.36 with an infinite upper bound. Every sector type contributes, because every slope is positive: banks

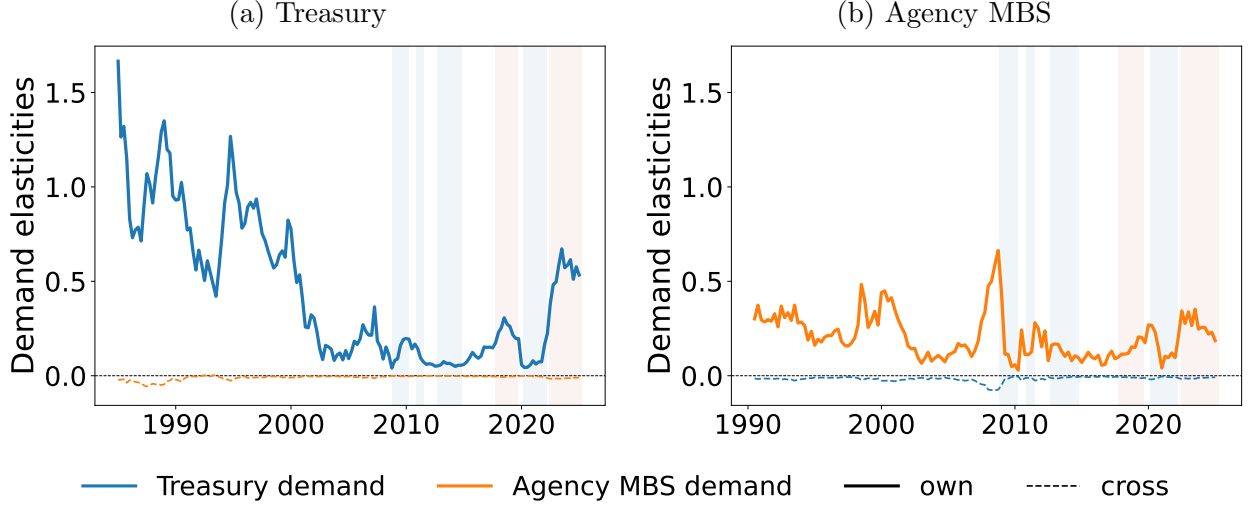
hold 33 percent of the private float, mutual funds 17 percent, foreign investors 14 percent, and the household residual 11 percent, and none of them subtracts. The contrast with the Treasury market is the paper’s central asymmetry in estimated form. Treasury absorption rests on two sectors holding less than a third of the float, and on one of them for four-fifths of the slope; agency MBS absorption is spread across all eight sector types.

The two markets are also linked, and the direction matters for what follows. Because the MBS block prices the spread, the level of the MBS yield is inherited from the Treasury market: a supply shift in Treasuries passes into mortgage yields one for one, and the agency MBS market determines only the wedge on top. The well-identified part of the MBS result is therefore the spread, and the level component carries the Treasury market’s wider uncertainty with it.

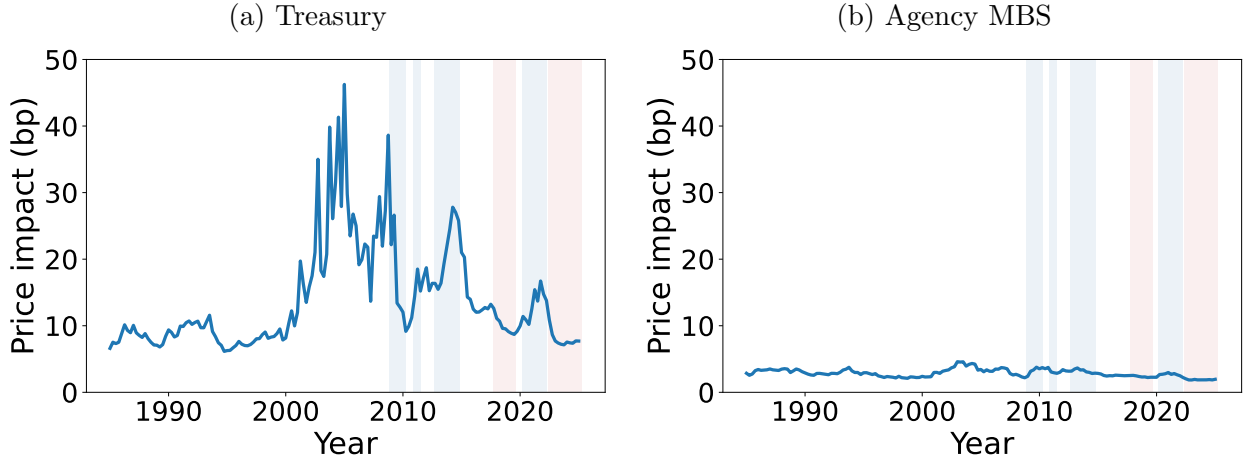
The system also implies the full set of own- and cross-price elasticities, and Panel A of Figure 3 plots them. Each of its two panels fixes a one-percent rise in one market’s price and traces demand in both. The own elasticities are sizable and move with the price level: Treasury demand responds by 0.45 percent per percent on average and by 0.53 percent at the 2025Q1 composition, falling through the low-yield decade and recovering after 2022, because the elasticity carries the yield level that the aggregate demand slope does not. The agency MBS entries are elasticities with respect to the spread, which averages 0.56 percentage points against a 4.4 point Treasury yield, so their smaller size is the price level and not a weaker response; the underlying semi-elasticity is four times the Treasury one. The cross elasticities sit near zero throughout, an order of magnitude below the own elasticities, because each scales with the share of the shocked asset in the portfolio and no bond is a large share of any sector’s balance sheet. Substitution on the demand side therefore runs between bonds and the outside asset, not between the two bonds. The spillover the counterfactual reports is the general-equilibrium counterpart, in which each market re-clears, not direct substitution. Appendix Figure A.7 shows the same object by sector type and Appendix Table A.12 reports the matrix.

Figure 3: **Bond Demand Elasticity and Implied Price Impact**

Panel A: **Market Demand Elasticities**



Panel B: **Implied Price Impact of 1% of Public Float**



Note: Panel A plots the market elasticities, the averages of the sector elasticities $\eta_{it}(n, n) = \beta_i(n)(1 - w_{it}(n)) y_t(n)$ and $\eta_{it}(m, n) = -\beta_i(m)w_{it}(m) y_t(m)$ weighted by holdings, $A_{it}w_{it}(n)$. Each subpanel raises that market's own $y_t(n)$ by one percent; the own response is solid, the cross response dashed. $y_t(n)$ is the ten-year yield for Treasuries and the option-adjusted spread for agency MBS, whose series begins in 1990Q3. Panel B plots the basis-point yield impact of removing 1 percent of the private float, $1/\sum_i s_{it}(n) \beta_i(n) (1 - w_{it}(n))$ (Equation 13), where $s_{it}(n)$ is sector i 's share of that float. It carries no $y_t(n)$ term and moves only with the composition of holdings. Each panel draws both markets on one scale: the agency MBS impact runs between 1.9 and 4.6 basis points, the Treasury impact between 6 and 46. Panel B shows point estimates; Table 5 gives the 2025Q1 Fieller interval. Both panels use the long-run slopes $\beta_i^{(4)}(n)$ of Table 4 with the cell models of Section 3.7, and floor at zero the household Treasury residual, which the Financial Accounts report as negative from 2004Q2 to 2008Q2 (Appendix D.2). Blue bands mark QE episodes, red bands QT episodes.

Table 5: **Aggregate Demand Slopes and Price Impact, 2025Q1**

Asset	Slope $\sum_i s_i \beta_i(n)(1 - w_i(n))$	t	Impact per 1% of float (bp)	95% Fieller band
Treasury	+0.130 (0.049)	2.7	7.7	[4.4, 28.9]
Agency MBS	+0.512 (0.140)	3.7	2.0	[1.3, 4.2]

Notes: The aggregate demand slope of Equation (13) at the 2025Q1 composition of holdings, with its delta-method standard error in parentheses (sector-type slopes treated as independent, and the two cells of Section 3.7 contributing none, since neither slope is estimated by its type’s regression), and the implied yield impact of removing 1 percent of the private float, $1/\text{slope}$, in basis points. The Fieller band inverts the 95 percent interval of the slope; it is unbounded above when that interval includes zero.

5 Counterfactual Simulation

5.1 Setup

The equilibrium asset yields are jointly determined by the estimated demand system in (8) and the market clearing condition in (14). To close the asset pricing model, the market for each asset n clears in each period t :

$$\sum_{i=1}^I w_{it}^*(\mathbf{y}_t^*, \mathbf{x}_t, \boldsymbol{\beta}_i, \boldsymbol{\varepsilon}_{it}) \cdot A_{it} = S_t(n; \mathbf{y}_t^*), \quad (18)$$

where $\mathbf{y}_t^*$ is the equilibrium yield vector, $\mathbf{x}_t$ is the asset-characteristics vector, $\boldsymbol{\beta}_i$ collects investor i ’s asset-specific demand slopes, $\boldsymbol{\varepsilon}_{it}$ are the latent demand shocks, and $S_t(n; \mathbf{y}_t^*)$ is the supply schedule of asset n , $S_t(n; \mathbf{y}) = f_n(\mathbf{y}; \bar{S}_t(n))$, which lets net issuance respond to the equilibrium yield; $\bar{S}_t(n)$, the free float at baseline yields, is the schedule’s shifter, and at baseline yields supply equals the observed float. Section 5.2 estimates the schedule’s reduced form. The equilibrium yield vector $\mathbf{y}_t^*$ is the implicit function

$$\mathbf{y}_t^* = \mathbf{g}(\mathbf{x}_t, \boldsymbol{\beta}, \boldsymbol{\varepsilon}_t, \mathbf{A}_t, \bar{\mathbf{S}}_t), \quad (19)$$

where $\mathbf{g}(\cdot)$ is the implicitly defined equilibrium mapping, which clears every market against the supply schedule, $\bar{\mathbf{S}}_t$ is the vector of baseline floats, and $\mathbf{A}_t$ is the vector of investor assets

under management.

The system is block-recursive in the two markets the Fed trades. Treasury demand responds to the ten-year yield and agency MBS demand to the option-adjusted spread (Section 3.6), so the Treasury market clears the level, the agency MBS market clears the spread, and the mortgage-relevant yield is the sum of the two,

$$\Delta y_t^*(\text{MBS}) = \underbrace{\Delta y_t^*(\text{Treasury})}_{\text{level, from the Treasury market}} + \underbrace{\Delta s_t^*}_{\text{spread, from the agency MBS market}}. \quad (20)$$

Supply is the one place the level re-enters the MBS block: origination responds to the mortgage rate, not to the spread, so the supply schedule of Equation (23) is evaluated at the reconstructed yield. Equation (20) is the decomposition the counterfactual reports throughout; a specification that prices MBS demand on the yield level cannot produce it, because its MBS market clears a price that already contains the Treasury level.

I impose clearing in those two markets.

I evaluate the impact of counterfactual monetary policy by comparing baseline and counterfactual equilibrium yields. Changes in the Fed’s asset holdings translate into changes in the baseline private free float $\bar{\mathbf{S}}_t$, and I measure them as the Fed’s transactions over the episode, the sum of its quarterly net purchases in the Z.1 flow tables, rather than as the change in the market value of its portfolio, which in 2022 mixes runoff with the revaluation of the securities it held. This one-for-one translation holds the fiscal path fixed: debt the Fed does not roll over is refinanced to the public at auction, and agency MBS paydowns are re-securitized, so runoff adds to the float exactly as outright sales would; any yield-induced change in gross issuance is what the supply schedule captures. Denoting the post-purchase baseline float by $\bar{\mathbf{S}}_t^{\text{CF}}$, the yield effect is

$$\Delta \mathbf{y}_t = \mathbf{g}(\mathbf{x}_t, \boldsymbol{\beta}, \boldsymbol{\varepsilon}_t, \mathbf{A}_t, \bar{\mathbf{S}}_t^{\text{CF}}) - \mathbf{g}(\mathbf{x}_t, \boldsymbol{\beta}, \boldsymbol{\varepsilon}_t, \mathbf{A}_t, \bar{\mathbf{S}}_t). \quad (21)$$

The logit system is nonlinear, and because some holders carry negative slopes it can have

several roots. I solve it by continuation, scaling the supply shift from zero to its full size in small steps and following the equilibrium that is continuously connected to the observed one. Where that path breaks, the shock is larger than the market’s absorbers can take at that date’s composition of holdings, and I report the first-order impact of Equation (13) instead, flagged as such.

5.2 Supply Response

The counterfactual lets bond supply respond to yields: when the counterfactual yield rises, issuers issue less (Fang and Xiao, 2025), attenuating the yield effect. This supply side is deliberately reduced-form, a first-order net-issuance elasticity rather than a structural model of issuer behavior, since the demand system is the paper’s structural focus. I estimate these elasticities from the same Flow-of-Funds data as the demand system. The identification mirrors the demand side. Estimating a demand slope requires supply-side variation in yields; estimating the supply (issuance) response requires *demand*-side variation, so the supply regressions use demand shifters, described below, rather than the demand system’s own instruments. The specification parallels the local projections of the demand system, with the response read at the same one-year horizon $h = 4$ as the demand slope $\beta^{(4)}$: the cumulative net issuance, measured from transactions as a percent of outstanding, through one year after a one-quarter yield change,

$$\sum_{s=t}^{t+4} \frac{F_s(n)}{S_{s-1}(n)} \times 100 = \kappa_i^{(4)} \Delta y_t(n) + \text{quarter fixed effects} + \tau t + u_t(n), \quad (22)$$

with $\Delta y_t(n)$ instrumented, quarter fixed effects absorbing the Treasury refunding cycle. The implied supply elasticity is $\psi_n = -\kappa_n^{(4)}$, the percent of outstanding withdrawn per percentage-point yield rise per year. The counterfactual’s supply schedule is its first-order form,

$$S_t(n; \mathbf{y}) = \bar{S}_t(n) [1 - \psi_n \Delta y_t(n)], \quad (23)$$

so relative to the baseline float $\bar{S}_t(n)$, issuers withdraw ψ_n percent of outstanding per percentage point that the counterfactual yield rises above its baseline value.

A supply curve has to be traced out by shifts in demand, just as a demand curve has to be traced out by shifts in supply. The coupon heteroskedasticity instrument identifies the demand side and so cannot identify supply: an instrument excluded from demand and correlated with the yield through supply cannot also be excluded from supply and correlated with the yield through demand. Used in the supply regression it returns a positive coefficient, +5.6 with a standard error of 5.8. I instrument instead with Japanese Ministry of Finance foreign-exchange intervention. When the Ministry sells yen and buys dollars it places the proceeds in Treasuries, so the shock originates in Japanese exchange-rate policy rather than in any decision of the US Treasury, and its first stage on the ten-year yield is 19.4. The sign flips: Treasury net issuance falls by 3.2 percent of outstanding per percentage point of yield per year, with a standard error of 6.1. The Fed's purchase-path revision is a demand shifter too, and I do not use it: it returns -27 , more than a quarter of the outstanding stock per percentage point, it survives a control for federal net borrowing, and it is identified off 2009 and 2020, when the purchase announcements and the fiscal expansions arrived together. Agency MBS issuance responds by -1.1 percent of outstanding per percentage point per year (standard error 3.4), identified by the [Drukker \(2025\)](#) price surprise. [Fang and Xiao \(2025\)](#) estimate the same aggregate object on the same kind of data and report 0.96 for Treasuries and 2.24 for agency MBS, but against the term spread and the MBS spread rather than the yields, so their numbers are a reference point and not a like-for-like benchmark; I do not carry them in the table. Measured from the change in market-value outstanding, every issuance elasticity is larger and more precise, because the market-value change contains the revaluation $-D \Delta y$ and so responds to yields by construction. The Treasury value is the one that matters for the headline numbers. At $\psi = 3.2$ the supply side absorbs a meaningful part of a Fed purchase. It absorbs the same part whatever the length of the program. One could scale ψ by the horizon, on the view that issuance keeps responding for as long as the yield

stays away from baseline, but the estimated path says the response is a stock adjustment rather than a flow. The Treasury $\kappa^{(h)}$ peaks at one year, -3.2 , and is back above zero by two, $+1.8$, where linear accumulation would have predicted -6.5 . Each episode therefore receives $\kappa^{(4)}$ once. The choice matters for the comparison of programs of similar size: QE2 and QE3 bought \$808 billion and \$817 billion of Treasuries, and under horizon scaling the ten-quarter program looked a third the size of the three-quarter one.

Table 6: **Net-Issuance Elasticities Estimated from the Flow of Funds**

Asset	One-year response $\kappa^{(4)}$		First-stage	Implied ψ
	Coefficient	(s.e.)	F	(%/pp/yr)
Treasury	-3.24	(6.09)	19.4	3.24
Agency MBS	-1.12	(3.42)	2.5	1.12

Notes: Instrumental-variables local projections of cumulative net issuance (transactions, percent of lagged outstanding) over the year following a one-quarter yield change, per asset, on 1985Q1–2025Q1; Newey–West standard errors in parentheses. The Treasury yield is instrumented by Japanese foreign-exchange intervention, a shift in foreign demand, and the agency MBS yield by the [Drukker \(2025\)](#) monetary surprise together with the transaction-based issuance interaction; the crisis indicator is a control. The two blocks are not equally well identified. The Treasury instrument reaches a first stage of 19.4; the agency MBS set reaches 2.5, and 3.9 with the Drukker surprise alone, which turns the point estimate positive, $+0.5$. A monetary surprise also moves mortgage origination directly, so it is not a pure demand shifter, and the Fed’s own agency MBS purchase announcements cannot replace it: they return $+26.7$, because the Fed announces mortgage purchases when housing is contracting and net issuance is falling. The agency MBS elasticity should therefore be read as weakly identified, and the counterfactual uses the smaller of the two point estimates. The supply schedule is written on the yield, not the spread: origination responds to the mortgage rate. The implied ψ is the elasticity used in the counterfactual’s supply response, set to zero where the coefficient is positive. *, **, and *** denote $|t|$ above 1.64, 1.96, and 2.58.

5.3 Assessment of Historical QE/QT Episodes

I apply the structural counterfactual to the four QE programs (QE1–QE4) and the two QT episodes (QT1, 2017–2019; QT2, 2022–2025). For each episode I hold investor characteristics, long-run demand slopes, and AUM at their observed values, let bond supply respond to yields through the net-issuance elasticities of Section 5.2, and ask how yields would differ had the

Fed not transacted. Table 7 reports the long-run effects, at the one-year slopes.

Four findings stand out. First, the Treasury effects are large, and they scale with the shock. QE1’s \$300 billion of Treasury purchases compressed the 10-year yield by 54 basis points, QE2’s \$808 billion by 103, QE3’s \$817 billion by 112, and QE4’s \$3.5 trillion by 186; scaled by the share of the private float removed (final column), the programs run from 8 to 12 basis points per 1 percent of float, net of the issuance the supply side returns to the market. They are large because the Treasury market’s absorber is thin: the household residual and the mutual funds, less than a third of the float between them, are the only sectors that take the other side, and the foreign, bank, and pension holdings that dominate the float do not move. QE2 and QE3 bought almost the same amount of Treasuries, \$808 billion against \$817 billion, and move the yield by 103 and 112 basis points. QE2 took a larger share of a smaller float, 12.2 percent against 9.2, but QE3 ran when the Treasury aggregate demand slope was lower, so each percent of float moved yields further.

Second, the agency MBS yield moves with the Treasury yield, and what the MBS market itself determines is the spread. Equation (20) splits each entry, and the split is the finding. QE1, the Fed’s only MBS-heavy program at \$1.2 trillion against \$300 billion of Treasuries, compressed the MBS yield by 108 basis points, of which 54 is the Treasury level and 54 is spread compression: this is the one program whose composition was mortgage-specific, and it is the only one with a large spread effect. QE3 and QE4 bought both markets in rough proportion and moved the spread by 31 basis points each; their MBS yield effects, 143 and 217 basis points, are mostly the Treasury level passing through. QE2 is the mirror image of QE1: the Fed bought \$808 billion of Treasuries while running off \$207 billion of MBS, which *widened* the spread by 12 basis points, so mortgage yields fell 90 basis points against the Treasury market’s 103 and homeowners received seven-eighths of the pass-through. Composition, not size, determines what QE does to mortgage borrowers.

The QE1 estimate is also where the demand system now meets the event-study record. Its 108 basis points sit at the top of the range between [Hancock and Passmore \(2011\)](#) (−85)

and [Krishnamurthy and Vissing-Jorgensen \(2011\)](#) (-107). The reconciliation is mechanical: an MBS demand curve written on the yield level makes the MBS market absorb the whole level shock by itself, and the elastic absorbers then deliver a small number; written on the spread, the level arrives from the Treasury market and only the wedge is absorbed locally.

Third, QE4 is the largest program because its shock was largest, not because demand was steeper. It removed a quarter of the private Treasury float, and scaled by that shock it moved yields by 7.7 basis points per 1 percent, the lowest of the QE programs. Two-thirds of its Treasury purchases came in the first half of 2020, when the Fed bought in weeks to restore market functioning ([Vissing-Jorgensen, 2021](#)); re-basing the counterfactual to the stock-QE phase that followed isolates a \$1.2 to \$1.5 trillion program that moved Treasury yields by 82 to 96 basis points (Appendix Table [A.11](#)), about 10 basis points per 1 percent, because the residual's share of the float was low in 2021.

Fourth, the QT episodes reverse the sign. QT1's \$373 billion of Treasury runoff and \$318 billion of MBS runoff raised the two yields by 23 and 31 basis points, 8 per 1 percent of float in each market, with 9 basis points of that coming from the spread. QT2's \$1.6 trillion of Treasury runoff raised the ten-year yield by 65 basis points and the mortgage yield by 76, of which 10 is spread. The treatment of the two cells in Section [3.7](#) matters for this episode: with money funds pooled into the mutual-fund type and the dealer cell carrying a free elasticity, the Treasury aggregate demand slope is small enough that no equilibrium connected to the observed one exists at QT2's size. With the two cells modelled, the nonlinear system clears at every episode. The message is the one Section [4.4](#) anticipated: at the current composition of the float the Treasury market's aggregate demand slope is small enough that a shock of QT2's size is at the edge of what a static demand system can price.

Per unit of supply removed, the Treasury estimates agree with the demand-system literature. Scaled by the shock they run 8 to 12 basis points per 1 percent of float, net of the issuance the supply side returns, against the 8 to 13 basis points per percentage point of the central bank's share that [Eren et al. \(2026\)](#) report and the 13 basis points per 1 per-

Table 7: **Yield Effects of Historical QE/QT Episodes**

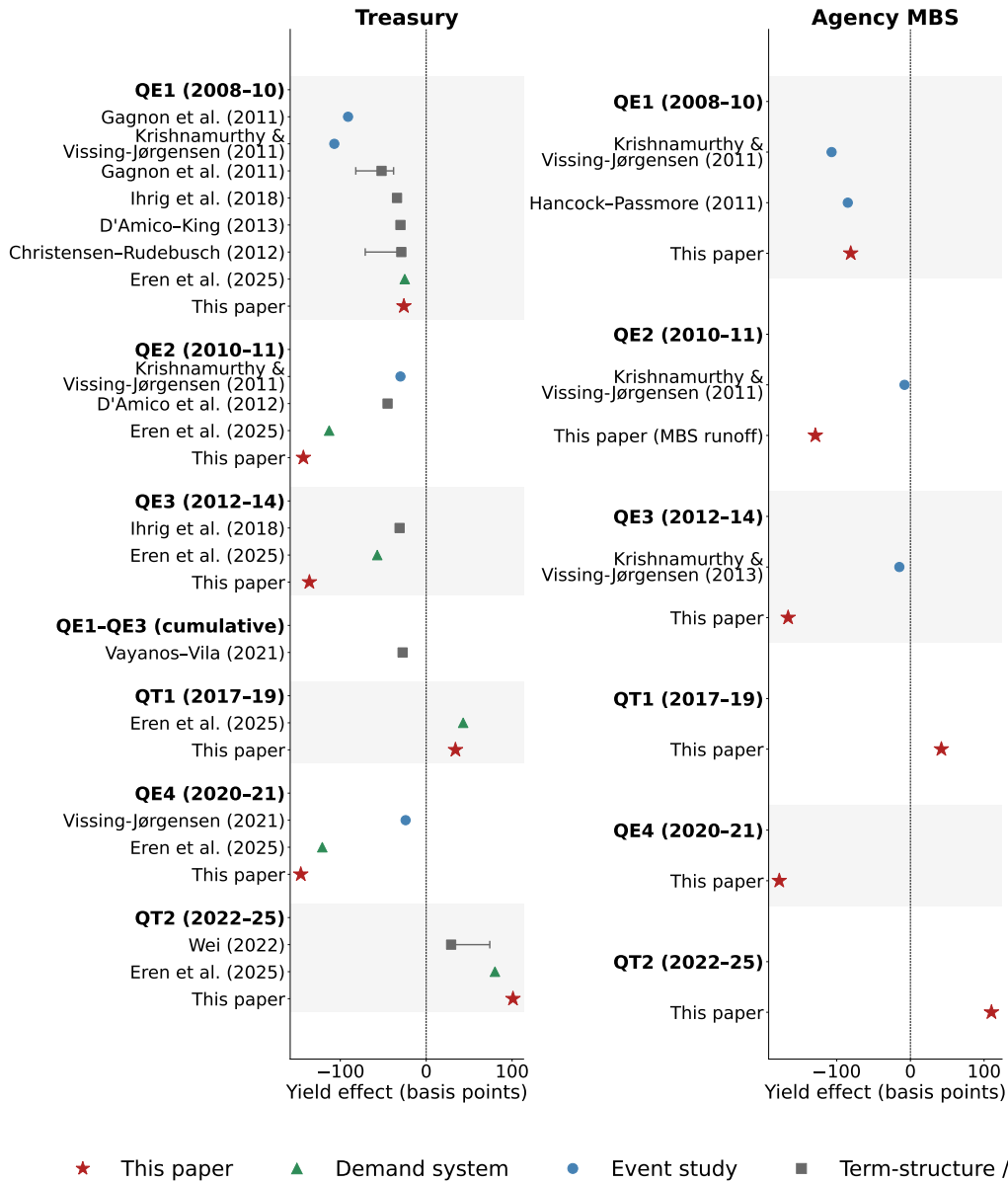
(a) Quantitative easing					
Episode	Asset	Purchases (\$B)	ΔY (bp)	of which spread (bp)	ΔY per 1% of float (bp)
QE1	Treasury	300	−54.4	—	10.4
2008Q4–2010Q1	Agency MBS	1,223	−108.4	−54.0	5.5
QE2	Treasury	808	−102.7	—	8.4
2010Q4–2011Q2	Agency MBS	−207	−90.2	12.4	28.1
QE3	Treasury	817	−112.4	—	12.2
2012Q3–2014Q4	Agency MBS	865	−143.0	−30.6	10.4
QE4	Treasury	3,516	−186.0	—	7.7
2020Q1–2021Q4	Agency MBS	1,238	−216.8	−30.8	14.5
(b) Quantitative tightening					
Episode	Asset	Runoff (\$B)	ΔY (bp)	of which spread (bp)	ΔY per 1% of float (bp)
QT1	Treasury	−373	22.7	—	7.8
2017Q4–2019Q3	Agency MBS	−318	31.2	8.5	8.1
QT2	Treasury	−1,622	65.4	—	7.7
2022Q2–2025Q1	Agency MBS	−549	75.6	10.3	13.3

Notes: Yield effects (basis points) of holding the Fed’s portfolio at its pre-episode level, from the market-clearing counterfactual of Section 5. The counterfactual shifts the baseline float by the Fed’s transactions over the episode, $\bar{\mathbf{S}}_t^{\text{CF}} = \bar{\mathbf{S}}_t + \Delta \text{SOMA}_t$, and re-solves the equilibrium in both markets against the supply schedule of Equation (23), with the elasticities of Table 6. All columns are measured at the episode-end quarter. Because the agency MBS block is estimated on the option-adjusted spread, the agency MBS yield change decomposes as in Equation (20) into the Treasury level, which passes through one for one, and the spread the MBS market clears; the “of which spread” column is the second term, and its positive entry in QE2 reflects the Fed’s concurrent MBS runoff. The last column scales the impact by the supply shock, $|\Delta Y|$ per 1 percent of the private float removed or returned. The money-market-fund and broker-dealer cells are treated as Section 3.7 describes.

cent shift in demand of [Chaudhary et al. \(2026\)](#). Figure 4 places the estimates episode by episode. The two systems are close on QE2 (103 here against their 113), and mine reads QE4 and QE3 larger (186 against 121, and 112 against 57). On the QT side mine are smaller, 23 against 43 for QT1 and 65 against 80 for QT2. Announcement-window event studies ([Gagnon et al., 2011](#); [Krishnamurthy and Vissing-Jorgensen, 2011](#)) are larger for QE1 and smaller for the later, largely anticipated programs; they price only the surprise component of purchases, on impact, and include the signaling channel that the supply-only counterfactual excludes. QE1, which the post-2010 demand systems cannot reach, enters here at 54 basis points, between the calibrated preferred-habitat estimate of 25 to 30 for the cumulative QE1 to QE3 purchases ([Vayanos and Vila, 2021](#)) and the event-study range of 91 to 107. On the mortgage side the agreement is closer: QE1’s 108 basis points lie at the top of the announcement-window range for agency MBS.

Three caveats bound the interpretation, and they do not all run in the same direction. The counterfactual holds latent demand and AUM fixed and treats the supply change as permanent, so the long-run estimates overstate the effect if investors expected the Fed to unwind ([Jansen et al., 2024](#)); the demand slopes are identified from local variation, while the episodes shift supply well outside that range; and the aggregate Treasury slope has a wide band, so the per-unit impact of 7.7 basis points at the 2025Q1 composition carries a 95 percent Fieller interval of [4.4, 28.9] (Table 5), finite at both ends but wide at the top. Weak-instrument bias runs the same way: it pulls the absorbers’ slopes toward zero, which makes the yield effects, if anything, too large. The two components of the mortgage number are bounded differently. The spread component is the tightest object in the paper: 2.0 basis points per 1 percent of the MBS float, with a two-sided Fieller interval of [1.3, 4.2]. The level component is the Treasury number and inherits its band, so a reader who doubts the Treasury absorber should read the mortgage result as the spread effect plus whatever the Treasury market delivers.

Figure 4: Historical QE/QT Yield Effects: This Paper versus Prior Estimates



Note: Structural full-stock estimates of Table 7 (stars) against prior estimates of each QE/QT episode, for Treasuries (left) and agency MBS (right). Circles are announcement-window event studies, squares term-structure and time-series estimates, triangles demand-system estimates; horizontal bars are ranges reported in the source study. The Vayanos and Vila (2021) entry is a preferred-habitat calibration of the cumulative QE1–QE3 purchases and appears as its own group, and the QT2 Treasury entry for this paper is a first-order impact. Episode definitions differ slightly across studies.

5.4 Prospective QT Scenario Analysis

I next project the yield effects of a resumption of quantitative tightening. Fixing the 2025Q1 portfolio and running additional runoff at the QT2 Treasury:MBS ratio, I consider three scenarios: a mild \$400 billion, a baseline \$1 trillion, about a year and a half at the Fed’s 2024 runoff pace of \$172 billion per quarter, and an aggressive \$1.6 trillion. Table 8 reports the results. A further \$1 trillion raises 10-year Treasury yields by 22 basis points and agency MBS yields by 27, of which 5 basis points is spread widening and the rest is the Treasury level passing through; the mild scenario raises them by 9 and 10, the aggressive by 33 and 39. A back-of-the-envelope calculation conveys the real stakes. Applied to the roughly \$27 trillion of marketable federal debt as it reprices, the baseline Treasury rise adds about \$60 billion a year to federal interest costs, a recurring fiscal cost to weigh against the balance-sheet reduction. On the household side, agency MBS yields pass through nearly one for one to primary mortgage rates, so the baseline adds about \$72 a month to the payment on a newly originated \$400,000 thirty-year fixed-rate mortgage.⁴ The marginal Treasury impact is nearly constant, about 2 basis points per \$100 billion of additional runoff. The mortgage cost of quantitative tightening is therefore mostly a Treasury-market phenomenon: a runoff split 73:27 leaves the mortgage spread almost unchanged, and the homeowner pays for the level.

The Treasury number is conditional on the marginal buyer, and I report it as a band rather than a point. Two cells carry the Treasury aggregate demand slope, and neither is pinned down. The household residual’s slope of 0.73 supplies four-fifths of the aggregate at the 2025Q1 composition, and it is identified on the pre-2010 sample: in the 2010–2025 window the sector’s slope is -0.84 with a standard error of 0.82 (Section 4). The foreign sector carries a larger weight still in the aggregate demand slope, because it holds a third of the float, and its slope is zero under the paper’s instruments but 0.48 under the Fed

⁴Fiscal: $\$27 \text{ trillion} \times 22.4 \text{ basis points} \approx \$60 \text{ billion per year}$ once the stock has repriced. Mortgage: the monthly payment on a loan of size P at annual rate r is the standard annuity $M(r) = P(r/12) / [1 - (1 + r/12)^{-360}]$; with $P = \$400,000$ and $r = 6.5\%$, $M(r + 27\text{bp}) - M(r) \approx \72 per month.

Table 8: **Yield Effects of Prospective Quantitative Tightening**

(a) Baseline

Scenario	Asset	Runoff (\$B)	ΔY (bp)	of which spread (bp)	ΔY per 1% of float (bp)
Mild (\$400B)	Treasury	−292	9.4	—	6.2
	Agency MBS	−108	11.3	2.0	10.1
Baseline (\$1T)	Treasury	−730	22.4	—	5.9
	Agency MBS	−270	27.3	4.8	9.7
Aggressive (\$1.6T)	Treasury	−1,168	34.5	—	5.7
	Agency MBS	−432	42.2	7.6	9.4

(b) Sensitivity to Treasury slopes

Scenario	Asset	Runoff (\$B)	ΔY (bp)	of which spread (bp)	ΔY per 1% of float (bp)
Expansion slope	Treasury	−730	34.8	—	9.2
$\beta_{\text{HR}}^{\text{Treasury}}$: baseline $\rightarrow$ 0.34	Agency MBS	−270	39.4	4.6	14.0
Recession slope	Treasury	−730	6.2	—	1.6
$\beta_{\text{HR}}^{\text{Treasury}}$: baseline $\rightarrow$ 3.58	Agency MBS	−270	11.3	5.2	4.0
Lower AR bound	Treasury	−730	44.8	—	11.8
$\beta_{\text{HR}}^{\text{Treasury}}$: baseline $\rightarrow$ 0.18	Agency MBS	−270	49.2	4.4	17.5
Higher foreign slope	Treasury	−730	12.9	—	3.4
$\beta_{\text{Foreign}}^{\text{Treasury}}$: baseline $\rightarrow$ 0.48	Agency MBS	−270	18.4	5.5	6.6
Two-year slopes	Treasury	−730	13.6	—	3.6
all slopes: $\beta^{(4)} \rightarrow \beta^{(8)}$	Agency MBS	−270	20.0	6.4	7.1

Notes: Panel (a) projects the yield uplift, in basis points, from runoff beyond the 2025Q1 balance sheet, fixing the 2025Q1 portfolio and solving the nonlinear clearing system in both markets with $\bar{\mathbf{S}}_t^{\text{CF}} = \bar{\mathbf{S}}_t + \Delta \text{SOMA}_t$ as the float after the runoff, split 73:27 as in QT2. Supply responds through the elasticities of Table 6. The impacts are long-run equilibrium, at the pooled household Treasury slope and the baseline foreign slope.

Panel (b) re-solves the \$1 trillion row of Panel (a) under alternative slopes, in the same layout and with the same supply response; each row label gives the slope assumed, with every other slope at its baseline value. The first three move the household residual's slope to the expansion and recession values of the cycle specification (Appendix D.7) and to the lower edge of its Anderson–Rubin set. The fourth moves the foreign slope from its baseline estimate of zero to 0.48, the value the purchase-path revision delivers alone (Appendix Table A.19). The last replaces every one-year slope with its two-year counterpart, $\beta^{(8)}$, leaving the supply response unchanged. The two cells of Section 3.7 keep their models.

purchase-path revision alone (Appendix Table A.19). Panel (b) of Table 8 re-solves the baseline scenario across both.

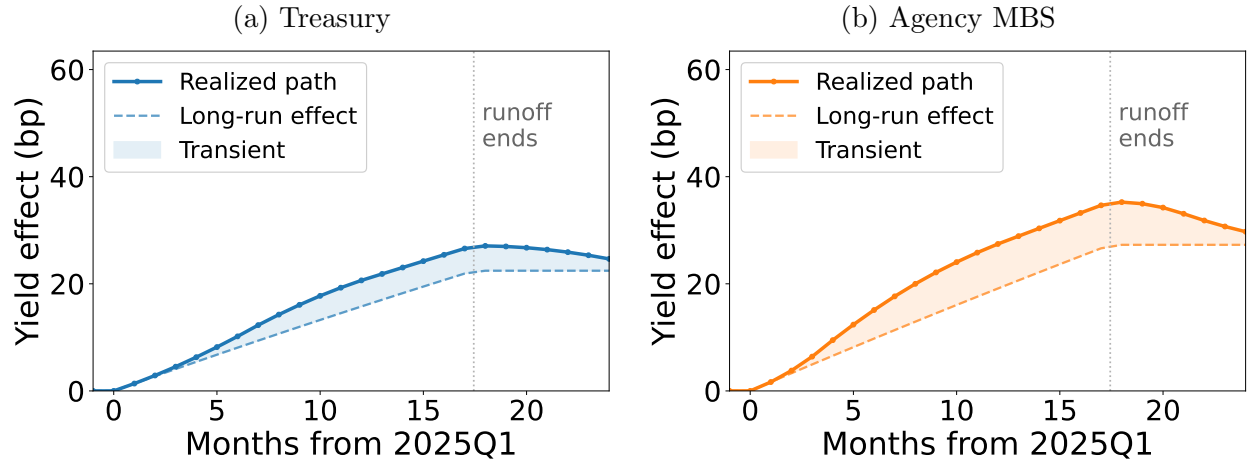
The household dimension is the one that can make the effect large, and the business cycle is what moves it. The residual sector's slope differs across states of the cycle more than any other cell's: 0.34 in expansions against 3.58 in NBER recessions, a difference significant at the 10 percent level ($p = 0.064$) (Appendix D.7). A runoff carried out in an expansion therefore raises the ten-year yield by 35 basis points and the same runoff in a recession by 6, against 22 at the pooled slope. The statistical band runs the same way: at the lower edge of the slope's Anderson–Rubin set, 0.18, the effect is 45 basis points. That edge has an economic reading, not only a statistical one. The residual sector is now mostly the hedge funds running the cash–futures basis trade, which hold 122 percent of its coupon Treasuries on average since 2013 (Section 4.2), while the slope applied to it is identified on the pre-2010 sample, when the same residual was households and nonprofits. A low household slope is the case in which the basis trade absorbs less readily than the buy-and-hold investor it replaced. The foreign dimension runs the other way: with foreign investors at 0.48 the same runoff raises Treasury yields by 13 basis points rather than 22. So does the horizon: several absorbers keep building through the second year (Section 4.3), and with every slope read at two years the runoff raises Treasury yields by 14 basis points and mortgage yields by 20. The honest range for a \$1 trillion runoff is therefore 6 to 45 basis points on the ten-year yield, every value of it a genuine equilibrium, and which end applies turns on two sector slopes that four decades of quarterly data do not separate. The Fieller band on the first-order impact tells the same story from the aggregate side: \$1 trillion is 3.8 percent of the Treasury float, and before any supply response the impact per 1 percent is 7.7 basis points with a 95 percent interval of [4.4, 28.9].

The mortgage number splits along the same seam, and the split is what makes it readable. The spread component barely moves across the band, 4 to 7 basis points in every row of Panel (b), because the MBS absorbers are identified in both windows and none of them

is the residual. Its own interval is two-sided, $[1.3, 4.2]$ basis points per 1 percent of the MBS float. The level component is the Treasury number, so the mortgage yield rises by 27 basis points at the pooled slope, 11 at the recession slope, 39 at the expansion slope, and 49 at the lower Anderson–Rubin edge. The honest summary is that the mortgage-specific piece of quantitative tightening is small and tightly estimated, while the mortgage cost of quantitative tightening is large and uncertain, because it is the Treasury market’s number in disguise, and it rests on the assumption that the sector which absorbed Treasuries before 2010, the household residual and the hedge funds it now contains, will absorb them again.

Figure 5 projects the resulting yield path, combining the runoff pace with the estimated rebalancing dynamics. Each quarter’s incremental runoff first lands on the impact demand curve: re-solving the \$1 trillion scenario horizon by horizon as the absorbers travel from $\beta^{(0)}$ to $\beta^{(4)}$, the impact quarter has no equilibrium at all, because the agency MBS aggregate slope is negative on impact and the Treasury slope is near zero, and the path peaks one to two quarters after the shock at about 1.6 times the long-run Treasury effect and 1.9 times the long-run mortgage effect, then decays to the long run within a year as slow capital arrives. Cumulating these vintages at the actual pace, Treasury yields rise by about one and a half basis points per month while the runoff proceeds, peak at 27 basis points when it ends after eighteen months, and are still 25 two years out, on their way to the long-run 22; the mortgage path peaks at 35 and is 30 two years out, against a long-run 27. The projection says most of the yield effect of a resumed QT arrives *during* the runoff, roughly in proportion to the stock withdrawn, and that the gradual pace limits the transient: at any time only the most recent increments are mid-adjustment, so the path peaks only a few basis points above its long-run component. The impact overshoot and its decay are the price-impact counterpart of the horizon duality [van der Beck \(2026\)](#) documents for equities, and the adjustment dynamics of [Greenwood et al. \(2018\)](#), in which a supply shock is absorbed first by local specialists and only gradually by cross-market capital.

Figure 5: **Projected Yield Path of a Resumed QT**



Note: Projected yield effects of \$1 trillion of further runoff from the 2025Q1 portfolio at the 2024 pace (\$172 billion per quarter, rendered monthly at \$57 billion per month), split between Treasuries and agency MBS at the QT2 ratio. Each month's incremental runoff enters at its impact effect and converges to its long-run effect along the adjustment path obtained by re-solving the scenario's equilibrium horizon by horizon as the absorbers travel from $\beta^{(0)}$ to $\beta^{(4)}$; the long-run component (dashed) accumulates the equilibrium effects of the runoff to date, with the supply response of Section 5.2 applied once, as in Table 8. Both panels share one vertical scale; the agency MBS panel is the yield, Treasury level plus spread. The shaded gap is the transient carried by the increments still mid-adjustment, which peaks when the runoff ends and dissipates as slow capital arrives. The vertical dotted line marks the end of the runoff.

6 Conclusion

The balance sheet will be run down again, and the framework here is built for the questions a resumed runoff will raise: who takes the other side, how far yields rise, and how the pace shapes the path. Its answers rest on measuring demand from what investors actually trade. Measured that way, most investors are inelastic, the few that absorb supply build their positions slowly, and the main Treasury absorber is the household residual, whose slope is identified mostly before 2010. Across the programs from QE1 to QT2, Treasury yields moved by 8 to 12 basis points per percent of the float, and the Treasury market set the level of mortgage yields while the MBS market set only the spread. Three caveats bound the answers. The Financial Accounts report no within-class maturity detail, so the system prices substitution across asset classes rather than along the curve; the supply side is deliberately reduced-form; and the demand slopes are identified from local variation, while the largest programs shift supply well outside it. Within those bounds, a further \$1 trillion of runoff from the 2025Q1 balance sheet widens the mortgage spread by a small and precisely estimated 5 basis points and raises Treasury yields by about 22. The Treasury number roughly doubles if the household residual absorbs Treasuries less readily than it once did, and it falls by a third if the absorbers are given two years rather than one to rebalance; the gradual pace of the runoff keeps the transient above the long-run effect small.

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A A BLP Interpretation of Sector-Level Asset Demand

In this appendix, I introduce a within-sector investor-level utility representation that provides a BLP-style interpretation of the demand estimation in the paper. Intuitively, each sector aggregates many underlying institutions (e.g., the U.S. banking sector includes large banks such as JPMorgan Chase and Bank of America), but the empirical analysis uses sector-level portfolios rather than institution-level holdings. For each sector i and within-sector investor h , the indirect utility from holding asset n at time t is

$$u_{hit}(n) = \beta_i y_t(n) + \gamma'_i x_t(n) + \xi_t(n) + v_{hit}(n), \quad (24)$$

while the outside asset $n = 0$ yields

$$u_{hit}(0) = \xi_t(0) + v_{hit}(0). \quad (25)$$

Here, β_i and γ_i denote sector-specific preference parameters that are common to all investors h in sector i . The term $\xi_t(n)$ captures asset-time fixed effects that are common across sectors and investors, such as unobserved asset characteristics, aggregate risk premium, or market-wide valuation components. Finally, $v_{hit}(n)$ are i.i.d. Type-I extreme value shocks that generate idiosyncratic within-sector heterogeneity. Although random coefficient specifications are standard in the BLP literature, I restrict within-sector heterogeneity to idiosyncratic shocks because sector-level data do not contain sector-specific price variation required to identify a distribution of random coefficients. Put differently, each sector is analogous to a market in the standard BLP framework. However, all sectors observe the same asset prices in financial markets, which by construction provides insufficient price variation to identify within-sector heterogeneity.

Given the Type-I extreme value errors, the portfolio weight of sector i on asset n at time

t is

$$w_{it}(n) = \frac{\exp\{u_{iht}(n)\}}{\sum_{k=0}^N \exp\{u_{iht}(k)\}}. \quad (26)$$

Normalize the outside option by setting $\xi_t(0) = 0$ and subtract $u_{iht}(0)$ from all utilities.

Under this normalization, the portfolio weight can be written as in Equations (5) and (6):

$$w_{it}(n) = \frac{\delta_{it}(n)}{1 + \sum_{k=1}^N \delta_{it}(k)}.$$

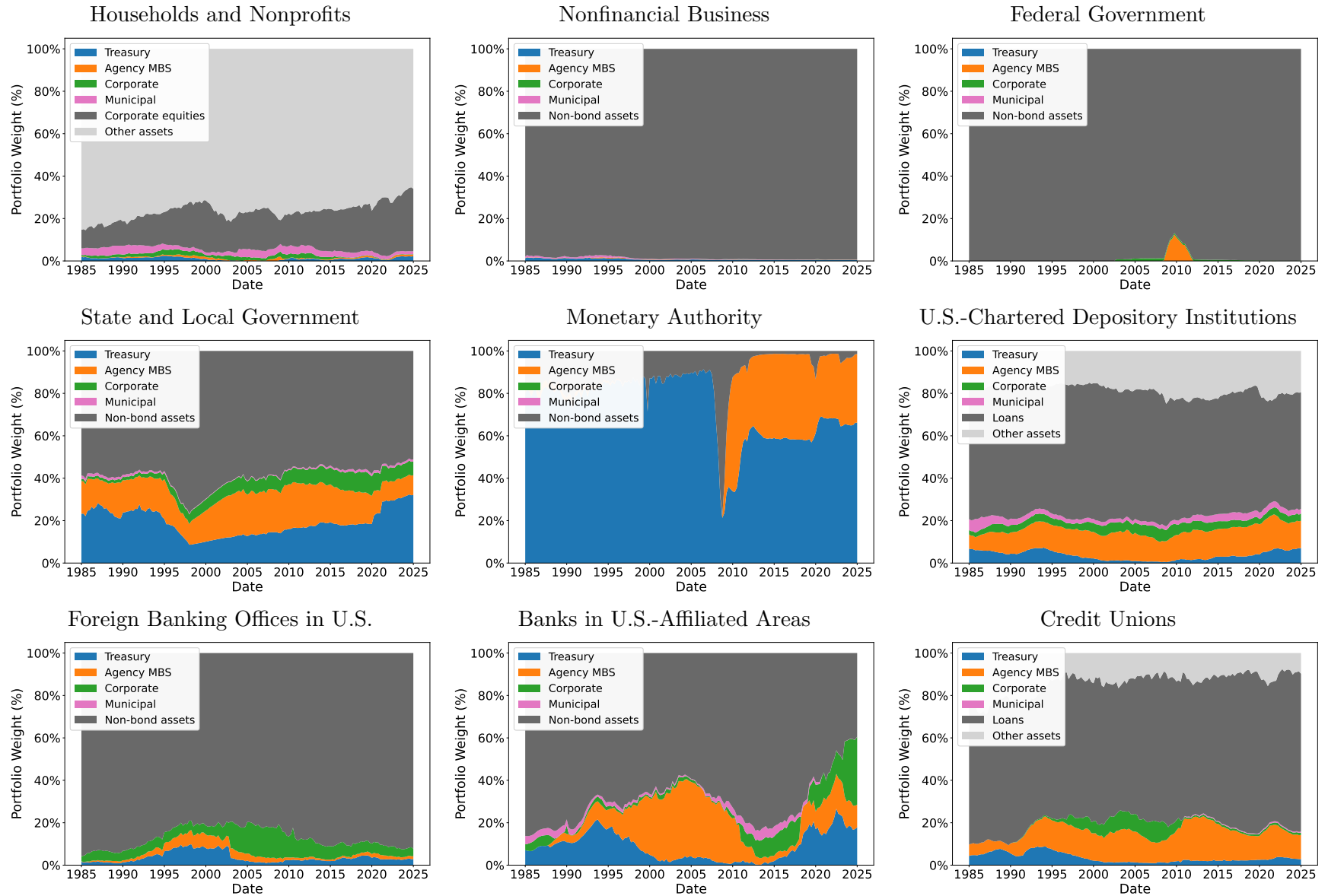
Here,

$$\delta_{it}(n) := \exp\left\{\beta_i y_t(n) + \gamma'_i x_t(n) + \xi_t(n)\right\} \epsilon_{it}(n),$$

where $\epsilon_{it}(n) := \exp\{v_{iht}(n) - v_{iht}(0)\}$.

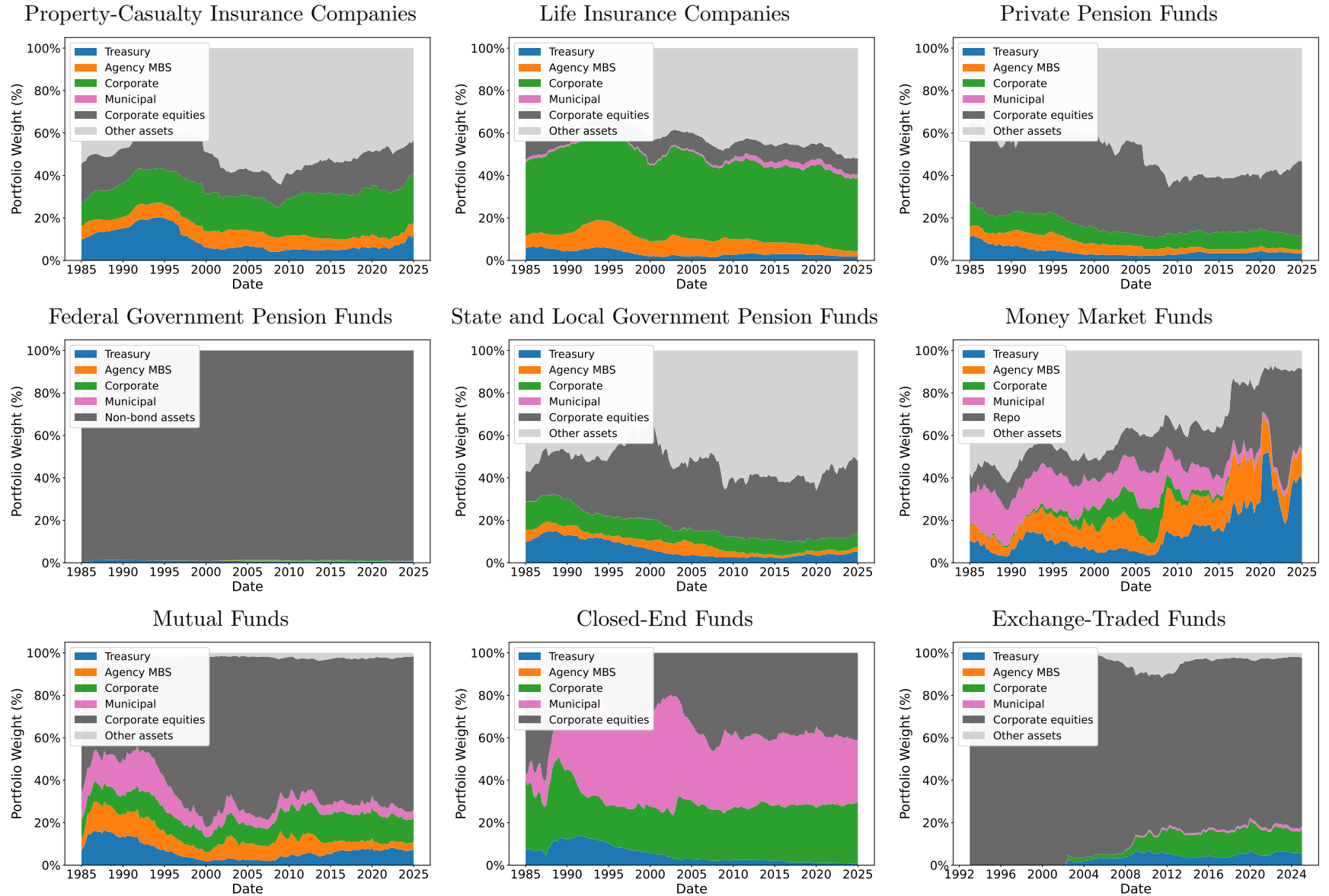
B Figures

Figure A.6: Portfolio Shares by Sector



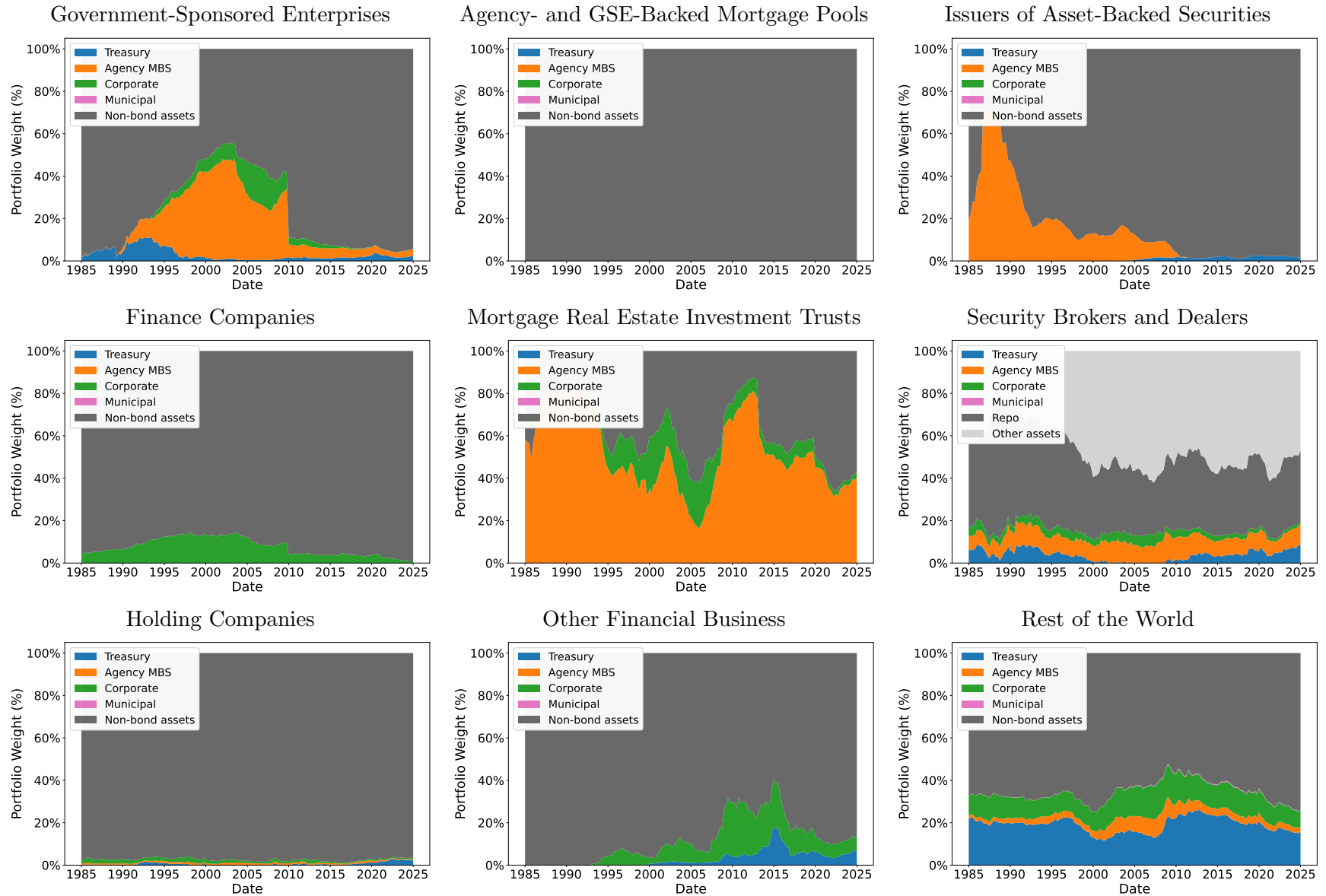
Notes: Stacked shares of total financial assets. Dark gray is the sector's own outside asset (Table 1); light gray is the remainder of the balance sheet, outside the demand system. For sectors whose outside asset is the total non-bond residual, the light-gray band is absent.

Figure A.6: Portfolio Shares by Sector (continued)



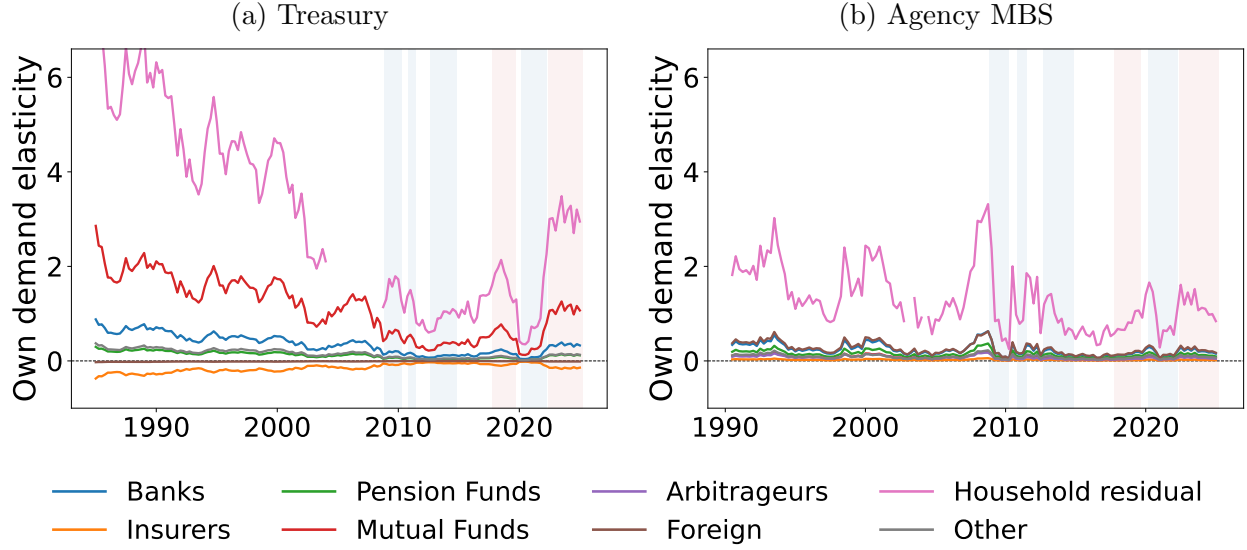
Notes: Stacked shares of total financial assets. Dark gray is the sector's own outside asset (Table 1); light gray is the remainder of the balance sheet, outside the demand system. For sectors whose outside asset is the total non-bond residual, the light-gray band is absent.

Figure A.6: Portfolio Shares by Sector (continued)



Notes: Stacked shares of total financial assets. Dark gray is the sector's own outside asset (Table 1); light gray is the remainder of the balance sheet, outside the demand system. For sectors whose outside asset is the total non-bond residual, the light-gray band is absent.

Figure A.7: Own Demand Elasticity by Sector Type



Note: Own elasticities $\eta_{it}(n, n) = \beta_i(n)(1 - w_{it}(n)) y_t(n)$, averaged within sector type with weights $A_{it}w_{it}(n)$. The figure shows logit cells only. The two cells of Section 3.7 that follow other models, money market funds and the dealer Treasury book, are left out of the type averages, so the mutual-fund Treasury line excludes money funds and the arbitrageurs have no Treasury line: the mortgage REITs in that type hold no Treasuries. Both panels are drawn on one fixed scale; the household residual's Treasury line leaves the frame before 1990.

C Tables

Table A.9: **Data Sources for Sector-Level Portfolio Holdings**

Type	Sector	Source
Banks	U.S.-Chartered Depositories	L.111
	Foreign Banking Offices	L.112
	Credit Unions	L.114
	Banks in Affiliated Areas	L.113
Insurers	Life Insurers	L.116
	Property-Casualty Insurers	L.115
Pension Funds	Private Pension Funds	L.118
	State & Local Govt. Retirement Funds	L.120
	Federal Govt. Retirement Funds	L.119
Mutual Funds	Mutual Funds	L.122
	Exchange-Traded Funds	L.124
	Money Market Funds	L.121
	Closed-End Funds	L.123
Arbitrageurs	Brokers and Dealers	L.130
	mREITs	L.129
Foreign	Foreign Official	CSLT
	Foreign Private	CSLT
Household Residual	Household Residual & Nonprofits	L.101
Other	Nonfinancial Business	L.102
	Government-Sponsored Enterprises	L.125
	Holding Companies	L.131
	State & Local Governments	L.107
	Federal Government	L.106
	Agency/GSE Mortgage Pools	L.126
	Finance Companies	L.128
	ABS Issuers	L.127
	Other Financial Business	L.132

Notes: Flow-of-Funds table codes (L. series) refer to the Financial Accounts of the United States (Federal Reserve Z.1); transactions come from the matching flow (FU) tables. The rest of the world is split into Foreign Official and Foreign Private holders over the full 1985–2025 sample using the Federal Reserve’s CSLT estimates of cross-border securities positions ([Bertaut and Judson, 2014](#)), which report monthly holdings and net transactions of U.S. Treasuries, agency bonds, corporate bonds, and equities by holder type; the Financial Accounts do not disaggregate foreign holdings by holder type.

Table A.10: **Data Sources for Asset Yields and Characteristics**

Asset	Variable	Source
Treasury	Yield	10-year constant-maturity yield, DGS10, FRED
	Time to maturity	YAS.MTY_YEARS, Bloomberg (LUATTRUU)
	Modified duration	YAS.RISK, Bloomberg (LUATTRUU)
	Liquidity	Author's computation from CRSP
	Issuance size	FL893061105, Z.1 Financial Accounts (L.210)
Agency MBS	Yield	10-year Treasury yield plus Bloomberg US MBS index OAS (LUMSTRUU)
	Time to maturity	WAL.TO_MATURITY, Bloomberg (LUMSTRUU)
	Modified duration	YAS.RISK, Bloomberg (LUMSTRUU)
	Liquidity	Author's computation from TRACE
	Issuance size	FL893061705, Z.1 Financial Accounts (L.211)

Notes: All yields are end-of-quarter. Maturity and duration are computed from the Bloomberg total-return index of each asset class (index tickers in parentheses); all indexes are unhedged and denominated in USD. Transactions and holdings by sector are from the Z.1 flow (FU) and level (FL/LM) tables. The liquidity measures do not enter the main specification.

Table A.11: **QE4 Decomposition: Market-Functioning versus Stock-QE Phase**

Phase	Fed Treasury purchases (\$B)	Treasury (bp)	Agency MBS (bp)	ΔY per 1% of float (bp)
QE4 full	3,516	−186	−217	7.7
stock QE (from 2020Q2)	1,462	−96	−114	9.5
stock QE (from 2020Q3)	1,212	−82	−99	9.8

Notes: QE4 (2019Q4–2021Q4) combined a market-functioning phase (2020Q1–Q2, \$2.3 trillion of Treasury purchases in the COVID crash) with a later stock-QE phase. Rows re-base the counterfactual baseline to isolate the stock phase, with the Fed’s transactions from the base quarter to 2021Q4 as the shock; the final column is the Treasury yield effect per 1 percent of float. The 2020H1 dislocation lies outside the stock-demand model, so the functioning phase is not reported as a stock effect. The phases are not additive, given the nonlinearity of the market-clearing map.

Table A.12: **Market Demand Elasticities**

Price shocked	Sample average		2025Q1	
	Treasury	Agency MBS	Treasury	Agency MBS
Treasury (10-year yield)	+0.452	−0.010	+0.533	−0.009
Agency MBS (option-adjusted spread)	−0.016	+0.208	−0.008	+0.186

Notes: The market elasticity matrix of Figure 3, averaged over the sample and evaluated at the 2025Q1 composition. Rows index the $y_t(n)$ that moves, columns the demand that responds; the diagonal is the own elasticity. A one-percent rise in the ten-year yield raises aggregate Treasury demand by 0.45 percent on average and by 0.53 percent at the 2025Q1 composition. The agency MBS entries are elasticities with respect to the option-adjusted spread, which averages 0.56 percentage points against a 4.4 point Treasury yield: the smaller number reflects the level of $y_t(n)$, not a weaker response, since the underlying semi-elasticity is 0.512 for agency MBS against 0.130 for Treasuries (Table 5). The cross elasticities are an order of magnitude smaller than the own elasticities because each scales with the share of the shocked asset in the portfolio, which is small for every sector.

D Specification Discussion

D.1 Outside-Asset Assignment, Sector by Sector

The demand system measures each sector’s bond demand relative to a single outside asset, the non-bond class the sector actually trades off against bonds (Table 1). This subsection records the rationale for each assignment, organized by outside asset.

Loans: U.S.-chartered depositories and credit unions. A depository’s marginal portfolio decision is between its securities book and its loan book: deposits fund either, and the securities portfolio serves as the buffer that absorbs deposit inflows when loan demand is weak. The QE literature documents exactly this margin: banks receiving reserve inflows under QE expand lending and adjust securities holdings in tandem ([Rodnyansky and Darmouni, 2017](#); [Kandrac and Schlusche, 2021](#); [Kurtzman et al., 2022](#)). Measuring bank bond demand against total non-bond assets would instead lump in reserves and interbank claims, which move with monetary policy mechanically rather than through portfolio choice.

Corporate equities: insurers, pension funds, mutual funds, and households. For the long-term investors the relevant margin is the stock–bond allocation. Insurers and pension funds manage fixed-income portfolios against equity allocations under asset–liability and return targets; mutual funds, together with ETFs and closed-end funds, reallocate between bond and equity exposures through both manager choice and investor flows; and household financial portfolios trade off directly held bonds against directly held equities. This is the classic rebalancing margin of the portfolio-balance channel: when QE compresses bond yields, these investors are pushed toward equities, and the outside asset is defined to capture precisely that substitution.

Repo: money market funds, broker-dealers, and hedge funds. Money funds allocate between short government paper and repo daily; repo is their actual alternative at the margin, not equities they cannot hold. Dealers likewise manage securities inventory against matched-book and reverse-repo positions, so repo is the financing margin along which their bond holdings expand and contract. Hedge funds enter the data in Treasuries only, where the Enhanced Financial Accounts data are informative and where the cash–futures basis trade concentrates their holdings. That trade is repo-funded: the fund buys the cash Treasury and finances it in the repo market, so repo, not equities or other bonds, is the margin along which its Treasury book expands and contracts, the same financing margin as dealers and money

funds. The main 26-sector panel keeps hedge funds inside the household residual, because a full carve-out is infeasible: Form PF includes offshore-domiciled funds, whose holdings the Z.1 assigns to the rest of the world, so subtracting the full hedge-fund book from the residual double-counts the offshore share and produces negative household holdings. The household residual’s Treasury slope in Table 4 is therefore the slope of a sector that, after 2010, contains the basis-trade book, and Section 4 reads it that way.

Non-bond U.S. securities: foreign investors. The rest of the world’s U.S. portfolio decision is between U.S. bonds and other U.S. securities, chiefly equities; the TIC-based holdings data are organized along exactly this line. Reserve managers, the dominant foreign Treasury holders, allocate between Treasuries, agencies, and other dollar assets, so the within-U.S. margin is the relevant one, not substitution back into home-currency assets.

Total non-bond (residual): the remaining sectors. For sectors without a single dominant alternative, foreign banking offices and banks in U.S.-affiliated areas (balance sheets dominated by interoffice claims), federal government retirement funds (holdings concentrated in special-issue Treasuries), mortgage REITs (levered MBS vehicles whose non-MBS assets are small), and the heterogeneous “Other” group, I use the residual of total financial assets net of every bond class the Financial Accounts report separately. Imposing a specific margin on these balance sheets would assert substitution behavior the data cannot support; the residual is agnostic.

Other bonds as a common outside asset. Corporate and municipal bonds are outside the choice set, so they can serve as one outside asset for every sector. That removes the cost Section 3 names: the slopes become comparable across sectors. It changes what the slope measures, however. The slope becomes substitution within fixed income, between Treasuries or agency MBS and credit, rather than substitution out of bonds, and a corporate yield that moves with the Treasury yield blunts it. I re-estimate the system with each sector’s

corporate and municipal book as its outside asset, keeping the instruments, controls, and cell models unchanged.

The outside asset changes who absorbs more than how much the market absorbs. The household residual's Treasury slope is 0.64 (standard error 0.28) against 0.73, though its Anderson–Rubin set now reaches -0.02 ; mutual funds fall from 0.28 to 0.11, and foreign investors rise from zero to 0.16. The Treasury aggregate demand slope rises from 0.130 to 0.160 ($t = 2.6$), so the impact per 1 percent of the float falls from 7.7 to 6.3 basis points; the agency MBS slope falls from 0.512 to 0.421, and its impact rises from 2.0 to 2.4. Historical Treasury effects shrink by a fifth to a third, to 6.3 to 7.9 basis points per percent of the float removed against 7.7 to 12.2 (QE1 -36 against -54 , QE4 -151 against -186). A \$1 trillion runoff raises Treasury yields by 19 basis points rather than 22 and mortgage yields by 26 rather than 27. The mortgage spread moves more: QE1 compresses it by 74 basis points against 54. The baseline keeps the non-bond margins, because the portfolio-balance channel of QE is the movement of duration out of the bond market, and a within-bond outside asset cannot see it.

D.2 Transactions versus Market Values

The Financial Accounts decompose the change in every level into transactions, revaluations, and other volume changes. The demand system uses the transactions (Section 2). This subsection documents why the alternative, the market-value share with an index-duration adjustment, does not measure active demand in these data, and what it does to the estimates.

The adjustment adds the revaluation back. For a bond position of duration $D_t(n)$, a yield change $\Delta y_t(n)$ (in percentage points) changes the position's market value by approximately $-D_t(n) \Delta y_t(n)/100$ even if the sector trades nothing, so the adjusted change

$$\Delta \tilde{q}_{it}(n) \equiv \Delta \ln \frac{w_{it}(n)}{w_{it}(0)} + D_t(n) \frac{\Delta y_t(n)}{100} \quad (27)$$

is meant to isolate active rebalancing. Eren et al. (2026) make this correction with a constant eight-year duration; the observed index duration is the natural alternative. The correction is right for a holder whose recorded level moves one for one with market prices at the index duration, and wrong otherwise. If a sector’s recorded level does not revalue, the differenced log share is already the active change, and adding $D \Delta y/100$ manufactures a positive response of $D/100$ per percentage point, a slope of 0.05 to 0.10 at the durations of the four indices, for a sector that traded nothing.

Table A.13 measures how much of the market revaluation each sector’s recorded level actually contains: the coefficient of the Financial Accounts’ own revaluation series on the index-implied revaluation $-D_t(n)\Delta y_t(n) H_{i,t-1}(n)/100$, cell by cell. The pass-through is a continuum. It is near one for the market-value holders, broker-dealers and open-end mutual funds; near zero for the Federal Government’s corporate holdings and for money market funds; and mostly between 0.1 and 0.5 for the depositories, insurers, and pension funds whose books mix amortized cost, statutory value, and market value. No sector-specific on-off correction reproduces this, and the index duration is the wrong duration for most of them.

Table A.14 shows what the measurement does to the demand slopes, separately from the instrument. It reports the common-slope sector-type long-run slopes $\beta^{(4)}$ on the full sample under the two measurements of the dependent variable, market-value shares with the adjustment (27) and transactions, and under two Treasury instruments, the market-value granular instrument and the heteroskedasticity instruments. The measurement, not the instrument, drives the results. With market-value shares, banks, insurers, and foreign investors carry small positive slopes (0.07, 0.12, 0.05 under the granular instrument), which would read as slow, weakly positive demand; with transactions, the same sectors are at zero or below under either instrument (-0.10 , -0.04 , -0.05 under the heteroskedasticity instruments), and the arbitrageurs’ large negative market-value slope shrinks to a fraction. Only the household residual is positive under transactions, and its slope rises when the

Table A.13: **Pass-Through of Market Revaluation into Recorded Levels**

Sector type	Sector	Treasury	Agency MBS
Arbitrageurs	Security Brokers and Dealers	1.50	0.81
Banks	Credit Unions	0.28	0.20
	Foreign Bank Offices	—	—
	Foreign Banking Offices in U.S.	—	—
	US Depositories	—	0.26
Foreign Investors	Rest of the World	0.36	0.27
Households	Households/Nonprofits	0.71	0.44
Insurers	Life Insurers	0.42	0.23
	Property-Casualty Insurers	0.32	0.15
Mutual Funds	Money Market Funds	-0.03	—
	Mutual Funds	0.83	—
Other	Federal Government	—	—
	Government-Sponsored Enterprises	-0.01	0.25
	Nonfinancial Business	0.33	—
	Other Financial Business	—	—
Pension Funds	Fed Pension Funds	0.27	0.54
	Private Pension Funds	0.40	0.33

Notes: For each sector and asset with a revaluation series in the Financial Accounts, the slope of the recorded quarterly revaluation on the index-implied revaluation of the lagged position, $-D_t(n) \Delta y_t(n) H_{i,t-1}(n)/100$, 1985Q1–2025Q1; a value of one means the recorded level moves with market prices at the index duration, a value of zero that it does not revalue. Cells are those for which the Financial Accounts publish the decomposition.

instrument is changed. The inert cluster was the adjustment.

The transaction slope and the market-value slope measure different objects, and they are related by an identity. Equation (4) measures the change in units held. The market-value share ratio changes by the same amount plus the relative price change, which is common to every holder of the asset and equals $-D_t(n)\Delta y_t(n)/100$ up to carry; regressing the transaction growth less the true revaluation on the yield change therefore gives a slope that differs from the transaction slope by $-D/100$, a uniform shift of about -0.05 across sectors. Under that normalization a holder that trades nothing has a slope of $-D/100$, not zero, and the household residual’s Treasury slope of about one is the one contrarian position in the market. Market clearing is written in units, and the Fed’s transactions shift units, so the counterfactual uses the transaction slope; the identity is why the choice of normalization does not affect it.

One cell is sensitive to how transaction growth is measured, and to how the level is treated.

Table A.14: Measurement versus Instrument: Sector-Type Long-Run Slopes

Sector type	Market-value shares, duration-adjusted		Transactions (FU)	
	Granular IV	Lewbel IV	Granular IV	Lewbel IV
Banks	+0.07 (0.18)	+0.08 (0.14)	-0.30 (0.09)	-0.10 (0.07)
Insurers	+0.12 (0.05)	+0.03 (0.04)	-0.03 (0.03)	-0.04 (0.03)
Pension Funds	-0.08 (0.07)	+0.05 (0.08)	-0.23 (0.07)	-0.09 (0.04)
Mutual Funds	-0.02 (0.09)	-0.39 (0.35)	-0.19 (0.06)	-0.10 (0.06)
Arbitrageurs	-0.58 (0.33)	-0.13 (0.27)	-0.05 (0.11)	-0.19 (0.16)
Foreign	+0.05 (0.05)	+0.02 (0.04)	-0.05 (0.03)	-0.05 (0.02)
Household Residual	+0.07 (0.53)	+0.29 (0.37)	+0.02 (0.17)	+0.46 (0.29)
Other	+0.00 (0.09)	-0.03 (0.06)	-0.15 (0.07)	-0.04 (0.08)

Notes: Sector-type long-run slopes $\beta^{(4)}$ from the pooled common-slope local projection of Equation (11) on 1985Q1–2025Q1, all sectors on the full sample, Newey–West standard errors in parentheses. Columns vary the dependent variable, the duration-adjusted market-value share change (27) or the transaction growth (4), and the Treasury instrument, the market-value granular instrument or the three-donor Lewbel heteroskedasticity instruments (all quarters); the agency MBS instruments are the SOMA transactions with the Drukker surprise throughout. The paper’s specification adds the crisis-robust construction and the asset-specific slopes of Table 4.

The Financial Accounts’ household Treasury position is a residual, total outstanding less every identified holder, and it is negative from 2004Q2 to 2008Q2, reaching $-\$318$ billion, about a tenth of the float, in 2007Q4. Wherever the paper weights sectors by their holdings, the aggregate demand slope, the market elasticities, and the counterfactual, I floor that residual at zero and let the float renormalize. A negative statistical residual would otherwise put a negative weight on the sector’s positive slope and turn the aggregate slope negative in six quarters, which is a measurement artifact and not demand; the reported counterfactuals are unaffected, because every quarter they solve at has a positive residual. Dealers’ negative Treasury positions are left as reported: a dealer can genuinely be net short, a residual cannot. Separately, the growth rate $\ln(1 + F/H_{-1})$ computed on that near-zero base is extreme in the surrounding quarters (its standard deviation is 0.70 over 2000–2008 against 0.09 before and 0.27 after). Without the 1/99 winsorization of Section 2 the household residual’s Treasury slope is 1.21 (standard error 0.36); with it, 0.73 (0.28); and with a 5/95 rule, 0.73 (0.26). The paper uses the 1/99 rule, fixed in advance for every cell, and the Treasury counterfactuals inherit this cell’s range.

D.3 Estimation Window

Table A.15 re-estimates the long-run slopes of Table 4 on the 2010Q1–2025Q1 window alone. The agency MBS absorbers keep their signs: the household residual at 1.03 (standard error 0.80) against 2.33 on the full sample, banks at 0.33 (0.23) against 0.48, and the arbitrageurs at 0.37 (0.26) against 0.15. The Treasury absorber does not: the household residual’s slope is -0.84 with a standard error of 0.82, and no other sector type’s Treasury slope is significant in the short window either. The Treasury counterfactuals of Section 5 therefore rest on the pre-2010 behavior of the residual sector, the dependence that Panel (b) of Table 8 quantifies.

Table A.15: **Long-Run Slopes on the 2010–2025 Window**

Sector type	Treasury (yield)	Agency MBS (spread)
Banks	+0.61 (0.35)	+0.33 (0.23)
Insurers	+0.01 (0.10)	-0.03 (0.09)
Pension Funds	+0.09 (0.13)	+0.05 (0.10)
Mutual Funds	+0.15 (0.12)	-0.24 (0.22)
Arbitrageurs	+0.07 (0.38)	+0.37 (0.26)
Foreign	-0.00 (0.06)	+0.02 (0.10)
Household Residual	-0.84 (0.82)	+1.03 (0.80)
Other	+0.03 (0.19)	-0.28 (0.44)

Notes: Long-run slopes $\beta^{(4)}$ from the local projections of Equation (11), estimated separately for each sector type and asset on 2010Q1–2025Q1 with the same instruments and controls as Table 4; Newey–West standard errors in parentheses. Dashes mark cells with too few observations.

D.4 Construction of the Purchase-Path Revisions

The instrument is built to isolate the unexpected part of the Fed’s balance-sheet policy. The Fed’s realized purchases are not a surprise: once a program is announced it runs on a published monthly schedule, so a quarter’s realized purchases are known to the market before the quarter begins. What the market could not anticipate is a change in that schedule, and the instrument is built from the communications that made one.

Events. I read the FOMC statements, minutes, and implementation notes between November 2008 and May 2024 and keep the twenty-one communications that changed the announced path of System Open Market Account purchases or runoff. Table A.16 lists them. Communications that restated the path, or changed only the timing of a program without changing the amount purchased over the following year, are kept in the record and coded zero.

Expected and unexpected purchases. For each event e and each asset n , agency MBS or Treasuries, let $P_e^{\text{new}}(n)$ be net purchases over the four quarters following the communication under the newly announced path, and $P_e^{\text{old}}(n)$ the same quantity under the path in force before it, which is the path announced at the previous event. The revision is

$$R_e(n) = P_e^{\text{new}}(n) - P_e^{\text{old}}(n). \quad (28)$$

The expected path is thus the one the Fed last announced: the market is assumed to expect the Fed to do what it last said it would do, and the whole change is the surprise. Nineteen of the twenty-one communications state an explicit total or monthly pace, so $R_e(n)$ follows from the announced amounts alone. The announcement of \$40 billion a month of agency MBS in September 2012, for example, is a revision of \$480 billion over the following year, and the \$600 billion of Treasury purchases in November 2010 is a revision of \$600 billion. Two communications name no amount and require inference: the June 2013 press conference, which said that purchases would begin to slow later in 2013 and end by mid-2014, and the

March 23, 2020 statement, which committed to purchases “in the amounts needed”. For the first I compute the reduction implied by that timetable relative to the pace then in force; for the second I use the purchases actually made through June 2020. Dropping both leaves the first stage essentially unchanged (Appendix Table A.18).

Timing and scaling. A change often becomes public before the FOMC statement that formalizes it, through a Jackson Hole speech, a press conference, or the minutes of an earlier meeting. I date each revision to the quarter in which the change first became public, taken from the documented prior communication; five events move, and the “Dated to” column of Table A.16 records them. Revisions dated to the same quarter are summed and scaled by the amount outstanding at the end of the previous quarter,

$$Z_t(n) = \frac{\sum_{e \in t} R_e(n)}{S_{t-1}(n)}, \quad (29)$$

so the instrument measures the surprise as a share of the market. The instrument is set to zero in the twelve crisis quarters of Section 3.6, which removes five communications, the November 2008 announcement, the September 2011 reinvestment decision, both March 2020 announcements, and the May 2022 runoff caps, each of which arrived during a flight to quality.

What the construction assumes. Treating the previously announced path as the expectation overstates the surprise when the market had priced a change from speeches or data that the record does not capture. Dating each change to its first public signal removes the documented cases. Any anticipation that remains makes part of the measured revision expected, which weakens the first stage rather than creating a spurious one. The exclusion restriction is that a revision affects sector demand only through the yield, conditional on the macroeconomic controls; the checks below show that the revisions do not respond to the previous quarter’s yields or macroeconomic conditions.

Table A.16: **FOMC Communications That Revised the Purchase Path**

FOMC date	Dated to	Agency MBS revision (\$bn)	Treasury revision (\$bn)	Coding	Communication
Nov 25, 2008	2008Q4 ^c	+500	0	explicit	LSAP announced: up to \$500bn agency MBS
Mar 18, 2009	2009Q1	+750	+300	explicit	MBS raised to \$1.25tn; \$300bn Treasuries added
Sep 23, 2009	2009Q3	0	0	explicit	Completion extended to 2010Q1 (timing only)
Aug 10, 2010	2010Q3	0	+253	explicit	MBS principal reinvested in Treasuries
Nov 03, 2010	2010Q3	0	+600	explicit	QE2: \$600bn Treasuries
Sep 21, 2011	2011Q3 ^c	+253	-253	explicit	MBS principal reinvested in MBS
Sep 13, 2012	2012Q3	+480	0	explicit	QE3: \$40bn/month agency MBS
Dec 12, 2012	2012Q4	0	+540	explicit	\$45bn/month Treasuries added
Jun 19, 2013	2013Q2	-200	-225	inferred	Taper to start in 2013, end mid-2014
Dec 18, 2013	2013Q4	+133	+175	explicit	Taper begins
Oct 29, 2014	2014Q4	0	0	explicit	Purchases end as scheduled
Jun 14, 2017	2017Q1	-72	-108	explicit	Runoff caps published
Sep 20, 2017	2017Q3	0	0	explicit	Runoff starts October 2017
Mar 20, 2019	2019Q1	0	+340	explicit	Treasury runoff to end September 2019
Mar 15, 2020	2020Q1 ^c	+437	+263	explicit	\$200bn MBS and \$500bn Treasuries
Mar 23, 2020	2020Q1 ^c	+560	+1,289	inferred	Purchases “in the amounts needed”
Jun 10, 2020	2020Q2 ^c	0	0	explicit	Pace codified, unchanged
Nov 03, 2021	2021Q3	-375	-750	explicit	Taper announced
Dec 15, 2021	2021Q4	-45	-90	explicit	Taper pace doubled
May 04, 2022	2022Q2 ^c	-368	-630	explicit	QT caps from June 2022
May 01, 2024	2024Q1	0	+420	explicit	Treasury cap cut from \$60bn to \$25bn/month

Notes: Revision to expected net purchases over the four quarters following each communication, relative to the previously announced path, in billions of dollars; negative values are reductions in purchases or increases in runoff. “Dated to” is the quarter in which the change first became public; ^c marks a crisis quarter, in which the instrument is set to zero. “Explicit” events state an amount or monthly pace; “inferred” events do not, and their revision is computed as described in the text.

D.5 Weak-Instrument Checks

Table A.17 reports three checks on the identification. The upper panel re-estimates the Treasury cells with all three heteroskedasticity donors, maturity, issuance, and coupon, in place of the single donor of the baseline. The three-donor block reaches a first stage of 6.4 against the baseline’s 7.7, because the added interactions carry no relevance and only dilute the statistic, and the long-run slopes are essentially unchanged: the household residual’s Treasury slope is 0.57 ([0.27, 1.56]) with three donors against 0.73 ([0.18, 1.91]) with the coupon donor alone. The bottom panel shows why the agency MBS block does not use heteroskedasticity instruments: adding them to the Fed’s transactions and the Drukker surprise raises the first stage to 17.3, but the overidentification test rejects the joint set for insurers, mutual funds, and the arbitrageurs, because the added relevance is the maturity interaction, and the index maturity of agency MBS moves with prepayment and hence with the yield. LIML and Fuller estimates coincide with two-stage least squares in every cell.

Table A.17: **Weak-Instrument Checks**

Asset	Sector type	Single donor (baseline)		Three donors	
		$\beta^{(4)}$	AR set	$\beta^{(4)}$	AR set
Treasury: first-stage F		7.7		6.4	
	Banks	+0.09 (0.11)	[-0.06, 0.36]	+0.03 (0.10)	$\emptyset$
	Insurers	-0.04 (0.06)	[-0.16, 0.14]	-0.05 (0.06)	[-0.20, 0.15]
	Pension Funds	+0.03 (0.04)	[-0.04, 0.11]	+0.03 (0.03)	[-0.04, 0.13]
	Mutual Funds	+0.28 (0.12)	[0.05, 0.65]	+0.28 (0.13)	[-0.04, 0.69]
	Arbitrageurs	-1.67 (1.14)	[-8.52, 0.16]	-1.44 (0.96)	[-9.09, -0.29] ^u
	Foreign	-0.00 (0.06)	[-0.10, 0.12]	-0.01 (0.07)	[-0.18, 0.14]
	Household Residual	+0.73 (0.28)	[0.18, 1.91]	+0.57 (0.25)	[0.27, 1.56]
	Other	+0.04 (0.14)	[-0.26, 0.34]	+0.05 (0.14)	[-0.28, 0.30]
Agency	MBS: SOMA + Drukker ($F = 3.1$) vs. +Lewbel ($F = 17.3$)	$\beta^{(4)}$	Sargan p	$\beta^{(4)}$	Sargan p
	Insurers	+0.60 (0.33)	0.39	+0.09 (0.08)	0.037
	Foreign	+0.77 (0.48)	0.09	+0.30 (0.18)	0.233
	Mutual Funds	+2.66 (1.44)	0.23	+0.14 (0.28)	0.000
	Arbitrageurs	+1.70 (1.18)	0.64	-0.03 (0.25)	0.018

Notes: Upper panel: long-run Treasury slopes $\beta^{(4)}$ with Newey–West standard errors and 95 percent Anderson–Rubin sets, under the paper’s instruments (the coupon heteroskedasticity instrument and the Treasury purchase-path revision) and with all three heteroskedasticity donors (maturity, issuance, coupon) in place of the coupon donor; $\emptyset$ marks an empty set and ^u a set reaching the edge of the grid. Lower panel: agency MBS slopes for the absorber cells under the Fed’s transactions and the Drukker (2025) surprise, and with the agency MBS heteroskedasticity instruments added, with the overidentification p -value of each set.

Crisis Quarters. The crisis quarters are defined by a criterion rather than by fit: the quarters of the flight-to-quality episodes documented in the literature, listed in Section 3.6. The August 2011 quarter is the one that matters for the estimates. It contains the U.S. downgrade and the escalation of the euro-area crisis, and it also contains the September 2011 decision to reinvest agency MBS principal in agency MBS, a large purchase revision in a quarter when the mortgage spread widened by 43 basis points. Its quarterly VIX, 30.6, exceeds that of 2008Q3 and of every 2022 quarter already in the set, and a mechanical VIX rule includes it under every threshold I tried. A mechanical VIX rule is not a good substitute for the criterion, however: it also adds the 1998–2003 and 2009 quarters and drops part of 2022, and under it the agency MBS instrument moves the Treasury yield as well, failing the cross-asset placebo, while the Treasury block loses more than half of its first stage.

The Menu of Candidate Instruments for Agency MBS. Table A.18 reports every candidate I examined for the agency MBS block, each run three ways: on the option-adjusted spread, which is the price the block estimates; on the Treasury yield, which is the cross-asset placebo an asset-specific supply shifter must fail to move; and on the MBS yield level, the price a yield specification would use. Read the first two column pairs together. The Fed’s realized transactions move the spread with an F of 3.1 but move the Treasury yield more (4.6), the wrong ordering for an MBS-specific instrument. The purchase-path revision over all quarters moves the spread at 5.3 and the Treasury yield at 12.0 ($t = -3.46$): in the crisis quarters the announcement and the flight to quality are the same event. Zeroed in those quarters the ordering is right and the gap is wide, 18.3 on the spread against 1.0 on the Treasury yield. Dropping the two judgment-coded events leaves this essentially unchanged (16.5 against 0.5), and dropping 2020–2025 leaves the spread first stage at 8.8.

The other candidates fail for identifiable reasons. Gross agency issuance at par is strongly correlated with the MBS yield but with a negative sign: issuance rises when rates fall because borrowers refinance, so the series traces the supply curve rather than shifting it, and it belongs

in the supply estimation of Section 5.2. Foreign official purchases of agency securities and the retained portfolios of the government-sponsored enterprises move with the MBS yield level (F of 6.6 and 10.2) but not with the spread, and the foreign series has the positive sign of buying after yields rise. The composition instrument, the MBS revision less the Treasury revision, has little power for the spread (2.4), and the LSAP factor of Swanson (2021) does not survive quarterly aggregation (0.3). The Treasury coupon instrument has no power inside the MBS block in the baseline control set, because the contemporaneous MBS index maturity moves with the rate level through prepayment and absorbs the level variation; entering that control with a one-quarter lag restores its relevance ($F = 19.3$), which is how the level-versus-spread restriction of Section 4 is tested.

Table A.18: **Candidate Instruments for the Agency MBS Block**

Candidate instrument	MBS spread		Treasury yield (placebo)		MBS yield	
	F	t	F	t	F	t
SOMA transactions + Drukker (2025) surprise	3.1	-2.05	4.6	+1.51	3.0	-2.02
SOMA transactions alone	4.2	-2.05	2.3	+1.51	4.1	-2.03
Drukker (2025) surprise alone	1.9	-1.37	6.7	-2.59	3.1	-1.77
Swanson (2021) LSAP factor	0.3	-0.51	0.6	+0.76	0.7	-0.83
Fed purchase-path revision, all quarters	5.3	-2.31	12.0	-3.46	9.0	-3.00
Fed purchase-path revision, crisis-robust	18.3	-4.27	1.0	-0.99	4.0	-1.99
composition revision, MBS less Treasury	2.4	-1.56	7.4	-2.71	0.4	-0.60
path revision, Treasury leg partialled out	5.4	-2.32	12.4	-3.52	2.8	-1.69
path revision + SOMA + Drukker	6.5	-2.39	8.3	-3.96	5.9	-2.96
Foreign official agency transactions (CSLT)	1.2	+1.08	0.4	+0.63	6.6	+2.56
GSE retained-portfolio transactions	0.9	-0.93	0.4	+0.64	10.2	-3.19
Treasury coupon Lewbel applied to agency MBS	2.3	+1.51	14.0	-3.74	3.4	+1.84

Notes: First-stage F and t statistics (Newey–West, four lags) of each candidate instrument on the quarterly change in the agency MBS option-adjusted spread, the ten-year Treasury yield (the cross-asset placebo), and the agency MBS yield level, 1990Q3–2025Q1, with the controls of the second stage. The bold row is the paper’s instrument.

Dating the Revisions: the Committee Signals Before It Votes. Dated to the FOMC statement that formalizes it, the Treasury leg of the purchase-path revisions has no first stage, with a positive sign ($t = +0.60$), meaning yields rise when the Fed announces that it will buy more. The event quarters look perverse. The Fed announced \$600 billion of

Treasury purchases in 2010Q4 and the ten-year yield rose 77 basis points over that quarter; it announced \$300 billion in 2009Q1 and the yield rose 46; it added \$45 billion a month in 2012Q4 and the yield rose 13.

The reason is anticipation. Regressing the quarterly change in the ten-year term premium on the statement-dated revision at t , $t+1$, and $t-1$ together, the contemporaneous coefficient is zero ($t = 0.18$) while the coefficient on the *next* quarter’s revision is negative ($t = -1.73$): the market prices part of a Treasury program before the Committee votes on it. The agency MBS leg is different, with a large contemporaneous coefficient ($t = -4.51$) and no lead ($t = -0.64$): what gets telegraphed is the size of a program, and the composition is the part that surprises.

I therefore date each revision to the quarter in which the change first became public, taken from the documented prior communication; Table A.16 lists the five events that move. Under this dating the Treasury revision loads on the ten-year term premium with $t = -2.57$ and on the yield with $t = -1.55$, against $+0.25$ and $+0.60$ under statement dating.

Two checks support using it. First, the revisions are not a reaction to the previous quarter: regressed on the prior quarter’s yield change, term-premium change, output growth, inflation, and volatility, the Treasury revision has an R^2 of 0.11, and instrumenting with only its unpredictable part leaves the first stage of the pair unchanged. Second, the Treasury revision moves the mortgage spread only with the opposite sign ($t = +2.17$), so it cannot be what identifies the agency MBS block. Paired with the coupon instrument the Treasury block has a joint F of 7.7, no sector type rejects the overidentifying restriction (minimum p of 0.30), and the Anderson–Rubin sets narrow.

Which Instrument Identifies the Treasury Block? The Treasury block uses two instruments, and a natural question is whether the internal one is needed at all. Table A.19 reports the block under each instrument alone, under the two external instruments together, and under the paper’s pair. A third external candidate appears there as well: Japanese

foreign-exchange intervention. When the Ministry of Finance sells yen and buys dollars it invests the proceeds in Treasuries, so an intervention shifts foreign demand for a reason that originates in Japanese exchange-rate policy. It is the strongest single instrument in the table, with a first stage of 16.4 and the sign a demand shift implies, and it moves the mortgage spread only weakly and with the opposite sign.

The instruments identify off different parts of the sample and different sectors. The coupon interaction has a first stage of 12.9 before 2009 and 0.2 after: the calm-period heteroskedasticity it exploits is a pre-crisis phenomenon. The external instruments are the mirror image, at 7.9 for the purchase-path revision and 6.7 for the Japanese intervention after 2009. They also disagree about who absorbs. The coupon instrument delivers the household residual at 0.76 and mutual funds at 0.28, the long-duration buy-and-hold absorbers; the external instruments deliver banks, at 0.51 with a set of $[0.05, 1.15]$, and leave the household residual at 0.77 with an unbounded set. That is a difference between periods rather than a conflict: the overidentification test rejects nowhere in the paper's pair.

The aggregate demand slope settles the question. At the 2025Q1 composition it is 0.130 (standard error 0.053) under the paper's pair and 0.132 (0.056) under the coupon instrument alone, both distinguishable from zero. Under the two external instruments together it is 0.156 with a standard error of 0.157, and under either of them alone it is not identified at all: the purchase-path revision alone gives 0.105 with a standard error of 0.311 and the Japanese intervention alone 0.191 with a standard error of 3.6, driven by the leveraged cells it cannot pin down. The household residual and the mutual funds carry most of the aggregate demand slope between them, and only the internal instrument identifies those two cells. What the comparison establishes is narrower and more useful than a ranking: the *level* of the Treasury aggregate demand slope is similar however it is identified, and the *identity* of the marginal buyer is not.

Table A.19: Treasury Demand by Instrument

Sector type	Coupon Lewbel		Fed path revision		Japan FX intervention		External only		Baseline	
	$\beta^{(4)}$	AR set	$\beta^{(4)}$	AR set	$\beta^{(4)}$	AR set	$\beta^{(4)}$	AR set	$\beta^{(4)}$	AR set
Banks	+0.07	[-0.13, 0.30]	+0.77	[-0.21, 2.30]	+0.48	[0.05, 1.01]	+0.51	[0.05, 1.15]	+0.09	[-0.06, 0.36]
Insurers	-0.04	[-0.14, 0.09]	+0.04	[-0.81, 0.67]	-0.00	[-0.27, 0.33]	+0.00	[-0.31, 0.37]	-0.04	[-0.16, 0.14]
Pension Funds	+0.03	[-0.02, 0.11]	-0.17	[-1.03, 0.44]	+0.08	[-0.07, 0.25]	+0.06	[-0.12, 0.26]	+0.03	[-0.04, 0.11]
Mutual Funds	+0.28	[0.05, 0.55]	+0.40	[-0.41, 1.37]	-0.09	[-0.79, 0.55]	-0.04	[-0.43, 0.67]	+0.28	[0.05, 0.65]
Arbitrageurs	-1.72	[-11.95, -0.18] ^u	-1.31	[-6.49, 1.39]	-18.73	[-1245.32, 1207.87] ^u	-1.41	[-10.66, 7.84] ^u	-1.67	[-8.52, 0.16]
Foreign	-0.01	[-0.14, 0.11]	+0.33	[-0.09, 2.06]	-0.02	[-0.20, 0.18]	+0.01	[-0.14, 0.21]	-0.00	[-0.10, 0.12]
Household residual	+0.76	[0.29, 1.59]	-0.17	[-5.47, 5.95]	+1.06	[-7.55, 9.66] ^u	+0.77	[-4.36, 8.96] ^u	+0.73	[0.18, 1.91]
Other	+0.05	[-0.20, 0.34]	-0.48	[-1.96, 0.86]	+0.26	[-0.40, 0.99]	+0.19	[-0.61, 0.79]	+0.04	[-0.26, 0.34]
First-stage F (ten-year yield)	14.0		3.5		16.4		8.8		7.7	
Aggregate demand slope, 2025Q1	+0.132 (0.056), t 2.3		+0.105 (0.311), t 0.3		+0.191 (3.566), t 0.1		+0.156 (0.157), t 1.0		+0.130 (0.053), t 2.5	

Notes: Long-run Treasury slopes $\beta^{(4)}$ from the local projections of Equation (11) on 1985Q1–2025Q1, by sector type, under five instrument sets: the crisis-robust coupon heteroskedasticity instrument alone; the first-signal Fed Treasury purchase-path revision alone; Japanese foreign-exchange intervention alone; the two external instruments together; and the coupon instrument with the purchase-path revision, which is the paper’s specification. The 95 percent Anderson–Rubin set appears beside each estimate; ^u marks a set reaching the edge of the grid. The last two rows report the first stage for the ten-year yield on the full sample and the implied 2025Q1 aggregate demand slope with a delta-method standard error, both computed under the paper’s treatment of the two misspecified cells (Section 3.7), so the last column reproduces the aggregate demand slope of Table 5. The standard error here treats the sector-type slopes as independent and is therefore slightly tighter than the one reported there.

Decomposing the Treasury Yield. A natural alternative is to price the Treasury block by the term premium, stripping the expectations component out of the yield the way the spread specification strips the Treasury level out of the mortgage yield. I use the decomposition of [Adrian et al. \(2013\)](#). The two instruments target different components: the coupon instrument’s power is in the expectations component, and the Treasury purchase-path revision’s is in the term premium. They pull in opposite directions, so the pair identifies the yield better than either component, with a first stage of 7.7 on the yield against 4.7 on the term premium.

Table A.20 reports both. The substance does not turn on the choice. The household residual and the mutual funds are the absorbers either way, at 0.73 and 0.28 on the yield and 1.47 and 0.77 on the term premium; the term-premium coefficients are larger because the term premium is the smaller and less variable price. The overidentifying restriction survives in every cell of the yield panel and in every cell of the term-premium panel but the arbitrageurs, at 0.06. I keep the yield in the main text because the counterfactual has to deliver basis points on Treasury borrowing costs and mortgage rates, and because the comparison with the QE literature is in yield space.

Table A.20: Treasury Demand Priced by the Yield and by the Term Premium

Sector type <i>first-stage F</i>	Ten-year yield (baseline)			Ten-year term premium		
	$\beta^{(4)}$	AR set 7.7	Sargan p	$\beta^{(4)}$	AR set 4.7	Sargan p
Banks	+0.09 (0.11)	[-0.06, 0.36]	0.30	+0.42 (0.29)	[-0.16, 1.24]	0.67
Insurers	-0.04 (0.06)	[-0.16, 0.14]	0.86	-0.09 (0.20)	[-0.49, 0.43]	0.64
Pension Funds	+0.03 (0.04)	[-0.04, 0.11]	0.44	+0.03 (0.13)	[-0.43, 0.44]	0.28
Mutual Funds	+0.28 (0.12)	[0.05, 0.65]	0.82	+0.77 (0.34)	[0.09, 1.45]	0.29
Arbitrageurs	-1.67 (1.14)	[-8.52, 0.16]	0.86	-1.74 (1.13)	[-2.42, 0.53]	0.06
Foreign	-0.00 (0.06)	[-0.10, 0.12]	0.36	+0.07 (0.19)	[-0.16, 0.81]	0.41
Household residual	+0.73 (0.28)	[0.18, 1.91]	0.60	+1.47 (0.67)	[0.80, 5.76]	0.21
Other	+0.04 (0.14)	[-0.26, 0.34]	0.43	-0.00 (0.39)	[-1.00, 0.77]	0.37

Notes: Long-run Treasury slopes $\beta^{(4)}$ from the local projections of Equation (11) on 1985Q1–2025Q1, by sector type, with the price variable in the left panel the ten-year yield, as in the main text, and in the right panel the ten-year term premium of Adrian et al. (2013). Both panels use the paper’s Treasury instruments, the crisis-robust coupon heteroskedasticity instrument and the first-signal Fed Treasury purchase-path revision, with Newey–West standard errors in parentheses, the 95 percent Anderson–Rubin set beside each estimate, and the Sargan overidentification p -value in the last column of each panel. The header row reports the joint first stage for each price. “ u ” marks a set reaching the edge of the grid.

The agency MBS purchase-path revision also loads on the Treasury term premium ($t = -2.13$) while leaving the expectations component untouched ($t = -0.64$), so it carries some duration content beyond the mortgage market. It does not move the ten-year yield that the Treasury block prices ($t = -0.94$), and the Treasury demand slopes are indistinguishable from zero for every sector type but two, so the substitution pressure this places on the agency MBS block is small. This is also why the spread specification matters on the mortgage side: news about growth and inflation moves the level of yields and leaves the mortgage spread alone, so the composition of an announcement is exogenous to the spread even when its size is not.

D.6 What the Household Residual Holds

The Treasury counterfactual rests on one cell, and the paper asserts that after 2010 that cell is mostly the hedge funds running the cash–futures basis trade. The Federal Reserve’s Enhanced Financial Accounts report hedge-fund balance sheets from Form PF quarterly since 2012Q4 and let the assertion be checked.

Over 2013Q1–2025Q1 hedge funds hold 122 percent of the residual’s coupon Treasuries on average, 99 percent at the median and 85 percent in the last quarter, and the two levels correlate at 0.93. In flows, the residual’s Treasury transactions regress on the quarterly change in hedge-fund holdings with a coefficient of 0.62 (Newey–West standard error 0.17, $t = 3.7$). The residual and the basis trade are the same position.

They cannot, however, be separated into two sectors and estimated jointly, which would be the natural way to add observations to the cell that carries the aggregate. The residual net of hedge funds is negative in 24 of the 49 quarters. The two series are not on the same basis: the Financial Accounts figure is a *net* residual, total outstanding less every identified holder, while the Form PF figure is a *gross* long financed by repo, and a repo-financed long is also in the Accounts as the lender’s holding, so differencing one from the other double-counts the financing leg.

What follows is a statement about interpretation rather than a correction. The slope the counterfactual applies to a hedge-fund position is identified where the same residual was households and nonprofits: the 2010–2025 window on its own gives -0.84 with a standard error of 0.82, an interval that contains every value the counterfactual could use. A levered arbitrageur and a buy-and-hold household are not obviously the same demand curve, and the lower Anderson–Rubin row of Panel (b) of Table 8, 45 basis points against 22, is the case in which the fund absorbs less readily than the household did.

D.7 The Household Residual over the Business Cycle

Every QE program ran in or just after a recession, and a demand system with one slope per cell prices them with a slope estimated mostly on expansions. This appendix adds one parameter, and only to the cell that needs it.

I screened all sixteen cells against three definitions of bad times, the NBER recession dates, quarters with the VIX in its top quintile, and quarters with negative real GDP growth,

estimating

$$y_{i,t}^{(4)} = \beta_i^G \Delta y_t + \delta_i (\Delta y_t \times \text{REC}_t) + \text{controls} + u_{it},$$

with each block’s instruments entered twice, as Z_t and $Z_t \times \text{REC}_t$, so both the level and the interaction are instrumented. Under the NBER dates two Treasury cells move. The household residual’s slope is $\beta^G = 0.34$ in expansions against $\beta^G + \delta = 3.58$ in recessions, with $\delta = 3.24$ and a standard error of 1.75, so $p = 0.064$; foreign investors move the other way, from +0.06 to −0.16 ($p = 0.04$), as reserve managers step back when the household residual steps in. The household residual’s agency MBS slope does not move, +0.69 on a standard error of 1.30, so the MBS block is left alone.

The household cell matters most because it carries 82 percent of the Treasury aggregate demand slope at the 2025Q1 composition. Table A.21 re-runs the historical episodes with each program’s household slope set to the slope of its own window, weighted by the share of quarters the NBER dates as a recession. The two recession programs fall into line with the literature: QE1 goes from −54 to −26 basis points against −25 in Eren et al. (2026), and QE4 from −186 to −146 against −121. QT1, which ran entirely in an expansion, goes from +23 to +34 against +43.

The restriction is not free, and the paper does not adopt it as the baseline. The expansion slope it implies, 0.34, is half the pooled 0.73 and carries a t of 1.18, so the programs that ran in expansions become larger and less certain: QE3 moves from −112 to −136 against −57 in the same comparison. Splitting sixteen recession quarters out of a hundred and fifty-nine buys a sharp answer for two episodes at the cost of precision everywhere else. What the exercise establishes is narrower and more useful: the asymmetry between QE and QT that Eren et al. (2026) document is in this system too, it lives in one cell, and it is the reason a symmetric specification overstates the recession-era programs.

Table A.21: **Historical Episodes with the Household Slope over the Cycle**

Episode	Recession quarters	ΔY Treasury (bp)		Household β used	Eren et al. (2026)
		Pooled slope	Cycle slope		
QE1 (2008Q4–2010Q1)	3 of 6	-54.4	-26.1	+1.96	-25
QE2 (2010Q4–2011Q2)	0 of 3	-102.7	-143.0	+0.34	-113
QE3 (2012Q3–2014Q4)	0 of 10	-112.4	-135.5	+0.34	-57
QT1 (2017Q4–2019Q3)	0 of 8	22.7	33.8	+0.34	43
QE4 (2020Q1–2021Q4)	2 of 8	-186.0	-146.4	+1.15	-121
QT2 (2022Q2–2025Q1)	0 of 12	65.4	100.6	+0.34	80

Notes: Treasury yield effects from the market-clearing counterfactual of Section 5. The pooled column is the paper’s specification, with the household residual at 0.73 in every quarter. The cycle column sets that cell to $0.34 + 3.24s$, where s is the share of the episode’s quarters the NBER dates as a recession; every other cell is unchanged. The last column is the corresponding estimate of Eren et al. (2026).

D.8 Nested Tests of the Slope Structure

The system estimates one slope per sector type and asset. Table A.22 asks whether the data need all sixteen, by minimum distance against two restrictions: one slope per type, which is the standard demand system, and a type factor times an asset loading. The first is rejected at the 10 percent level ($p = 0.06$) and the second is not rejected ($p = 0.29$), so the asset dimension matters but the data are content with a single loading ratio, 0.32 on the Treasury yield against 1.38 on the agency MBS spread. The counterfactual uses the unrestricted slopes.

Table A.22: **Nested Tests of the Slope Structure**

Restriction	Parameters	Distance χ^2 (df)	p
Common slope across assets, β_i	8	15.1 (8)	0.057
Proportional, $\beta_i \theta(n)$	9	8.5 (7)	0.294
Unrestricted, $\beta_i(n)$	16	0.0 (0)	—

Notes: Minimum-distance tests on the 16 long-run slopes of Table 4, treating the cells as independent with their Newey–West standard errors. “Common slope” restricts each sector type’s slope to be equal across Treasuries and agency MBS (8 parameters); “proportional” restricts it to a type factor times an asset loading, $\beta_i(n) = \beta_i \theta(n)$ (9 parameters, loadings normalized to unit mean square); “unrestricted” is the estimated matrix. The distance statistic is χ^2 with degrees of freedom equal to the number of restrictions.